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RESEARCH ARTICLE

SUPER: Smart User-Centric Popularity Exposure Reduction for Fair and Diverse Recommendations

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ABSTRACT We present *SUPER*, a novel approach to mitigating popularity bias in recommender systems by explicitly modeling user-specific preferences for item popularity. Unlike existing methods that apply uniform adjustments to reduce the dominance of popular items, *SUPER* leverages each individual's inclination toward mainstream versus niche content. By combining a Pareto-based item partitioning with personalized preference calibration, we incorporate both popular and tail subsets into the training and recommendation refinement processes. Experimental evaluations on multiple benchmark datasets demonstrate that *SUPER* substantially improves diversity and fairness while maintaining robust accuracy. Furthermore, our findings indicate that aligning the recommendations with each user's interaction-derived popularity tendency promotes a broader exposure to underrepresented items and produces more satisfying recommendation experiences. These insights underscore the potential of personalized debiasing strategies to promote equitable and context-sensitive content curation in modern recommendation systems.

INDEX TERMS Diversity, fairness, popularity bias problem, popularity debiasing, pre- and post-processing, user-centric recommendation.

I. INTRODUCTION

Recommender systems (RSs) have become indispensable components of modern digital platforms by guiding users toward content aligned with their interests in increasingly large information spaces. Leveraging data mining, machine learning, and artificial intelligence techniques, RSs identify patterns in user behavior and generate personalized suggestions that improve user satisfaction and engagement [1], [2]. Beyond classical applications such as e-commerce and media streaming, recommendation mechanisms also play an important role within expert systems, where intelligent decision-support frameworks rely on personalized guidance across various domains [3], [4]. Related challenges of preference imbalance and selective exposure have also been studied in adjacent intelligent systems domains, such as visual interaction and rehabilitation-oriented learning

frameworks, where user-centric modeling and imbalance-aware learning play a critical role [5], [6].

A persistent challenge faced by RSs is *popularity bias*, where a small subset of frequently consumed (*head*) items receives disproportionately high exposure, while the majority of less popular (*tail*) items remain under-recommended [7], [8], [9]. This “rich-get-richer” phenomenon reduces catalog coverage, limits meaningful discovery, and yields homogenized recommendation lists. Empirical studies have reported the prevalence of popularity bias across diverse domains, including movies [10], [11], education [12], tourism [13], books [14], and music [15], [16]. Beyond technical implications, popularity bias can also reinforce cultural uniformity and hinder the visibility of emerging or niche content [17]. Similar imbalance phenomena have also been studied in other data-driven domains, such as industrial computer-vision systems addressing information-density imbalance, highlighting that the challenge of uneven data and exposure distributions is a general

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learning problem rather than one unique to recommender systems [18].

To address these issues, a wide range of popularity-bias mitigation strategies has been proposed, including re-ranking approaches, fairness-aware objectives, and diversity-promoting constraints [10], [19], [20], [21], [22]. While effective at the aggregate level, many of these methods lack genuine *user-centricity*. By emphasizing global exposure balance, they may inadvertently disregard individual users' preferences, replacing popularity bias with a different form of misalignment between recommendations and user tastes. Recent research therefore highlights the importance of modeling user-specific popularity tendencies, aiming to reduce preference–exposure confounding factors such as visibility, promotion, and social influence [7], [23], [24], [25]. Recent advances further investigate causal modeling, adversarial robustness, and motivation-aware retrieval to better disentangle exposure from genuine user preference in modern recommender systems.

A. PROBLEM STATEMENT AND MOTIVATION

Despite notable progress, existing popularity-bias mitigation approaches often fail to capture the nuanced and evolving nature of users' preferences. A common implicit assumption is that interactions with popular items directly reflect a user's affinity for mainstream content. In practice, however, users may engage with trending or prominently displayed items due to convenience, discounts, or social exposure rather than genuine interest. Treating such interactions as stable preference signals can mislead RSs and produce recommendation lists that increasingly diverge from users' stable preference patterns (as observed in interactions). We note, however, that without impression logs it is generally not possible to identify causal user preferences from observational interaction data alone. Therefore, in this work, popularity inclination is modeled as a behavioral proxy used for calibration rather than as a ground-truth latent construct. For instance, consider a user with only a small number of interactions. Even within limited histories, users often exhibit observable differences in their consumption exposure patterns, some concentrate on widely consumed items, while others disproportionately engage with niche content. Although such signals do not constitute causal preference, they provide a practical behavioral basis for estimating a user-specific popularity-inclination proxy and calibrating head-tail exposure under standard offline recommendation settings.

Many existing methods rely on post-processing re-ranking to elevate tail items [10], [17], [20] or introduce fairness and diversity constraints to rebalance exposure [21], [22]. Although effective at improving catalog-level fairness, these techniques often operate on static or aggregate representations of user behavior. More recent work has moved toward user-aware modeling by considering popularity proportions within user profiles [26], [27] or incorporating contextual and temporal factors [7], [23], [24], [25]. Nevertheless, accurately

separating underlying preference signals from incidental exposure remains a key challenge.

These observations motivate the need for a user-centric popularity-debiasing framework that explicitly models each user's inclination toward popular versus tail content and uses this inclination to calibrate the final Top- N recommendation list. Such an approach aims to mitigate popularity bias without sacrificing personalization, enabling balanced, diverse, and preference-aligned recommendations.

Accordingly, the goal is not to replace existing debiasing paradigms, but to provide a structurally simple and model-agnostic mechanism that stabilizes popularity exposure at the user level. This perspective positions popularity calibration as a complementary control layer rather than a competing learning objective.

B. RESEARCH QUESTIONS AND CONTRIBUTIONS

This study introduces a smart user-centric popularity-debiasing framework that integrates user-specific popularity inclination into both the training and post-processing stages of recommendation generation. The framework is guided by the following research questions:

RQ1: How can we estimate a robust behavioral proxy of users' popularity inclination under interaction-only logs, while mitigating exposure–preference conflation?

RQ2: How can we integrate user-specific popularity tendencies into a hybrid recommendation framework that combines pre-processing strategies (training on popular and tail subsets separately) with a post-processing calibration step to ensure balanced and personalized top- N lists?

RQ3: How can we rigorously evaluate this smart method's effectiveness in mitigating popularity bias while maintaining or improving overall recommendation quality, diversity, and user satisfaction?

To address these questions, we make the following contributions:

- 1) **User-Centric Popularity Profiling:** We propose a refined technique that goes beyond simplistic measures of popularity engagement, employing the Pareto principle to identify and characterize a user's observed leaning toward popular or niche content.
- 2) **Hybrid Pre- and Post-Processing Framework:** Leveraging these user profiles, we introduce a hybrid approach where popular and tail item subsets are independently trained, then dynamically blended according to each user's detected popularity inclination.
- 3) **Comprehensive Evaluation Paradigm:** We develop a robust evaluation framework incorporating conventional accuracy metrics as well as measures of diversity, fairness, and user satisfaction. This provides a holistic understanding of the system's ability to mitigate popularity bias while aligning with individual preferences.

C. PAPER ORGANIZATION

The remainder of the paper is organized as follows. Section II reviews related work on popularity bias and its mitigation. Section III presents the proposed SUPER framework in detail. Section IV describes the experimental setup, datasets, and evaluation metrics. Section V reports and discusses the experimental results. Section VI outlines the limitations of the proposed approach, and Section VII concludes the paper and suggests avenues for future research.

II. RELATED WORK

Popularity-bias mitigation strategies in recommender systems are commonly categorized into *pre-processing*, *in-processing*, and *post-processing* approaches. Pre-processing methods aim to reduce imbalance in the training data before model learning, for example by separating head and tail items or reweighting interactions to amplify under-represented content [28], [29]. In-processing techniques modify the learning objective or model constraints to penalize popularity-correlated signals or explicitly optimize for diversity and fairness [10], [11], [19], [30], [31]. While effective, these approaches are often tightly coupled to specific model architectures and may require substantial redesign to generalize across recommenders.

Post-processing methods operate after a base recommender generates prediction scores and adjust the final ranked list to control exposure. Representative examples include inverse-popularity calibration [32], [33], diversification mechanisms such as xQuAD [20], and exposure-balancing formulations including Discrepancy Minimization [34] and FA*IR [35]. Due to their model-agnostic nature, post-processing approaches are particularly attractive for benchmarking and comparative evaluation across multiple backbone recommenders.

Among user-centric mitigation approaches, *Calibrated Popularity* [21] is closely related to our work, as it seeks to align the popularity distribution of recommendations with a user's historical consumption. However, a central challenge in such calibration lies in the fact that historical interactions may conflate preference with exposure. Treating all consumed items equally can overestimate affinity for popular content that was merely visible or promoted.

Some other studies therefore emphasize the need for behaviorally meaningful user profiling that better reflects genuine preference signals [7], [23], [24], [25]. Nevertheless, there remains a practical gap for methods that (i) operate consistently across diverse recommendation algorithms and (ii) calibrate popularity exposure at the individual user level while preserving ranking quality and beyond-accuracy objectives.

Recent research has increasingly focused on user-aware and optimization-driven strategies for mitigating popularity bias. Approaches such as *Popularity-aware Distributionally Robust Optimization (PDRO)* [36], *Personalized Popularity-Aware Calibration (PPAC)* [37], and *Causal Exposure-based*

Popularity Penalization (CausalEPP) [38] exemplify this trend by incorporating robustness, personalization, or causal reasoning into the debiasing process. While these methods improve exposure control through distributional optimization or adaptive calibration, they often rely on stronger modeling assumptions and tighter integration with the learning objective. Consequently, ensuring consistent rank-level alignment with user-specific popularity tendencies while preserving model-agnostic applicability remains an open challenge.

Beyond optimization-based and calibration-driven methods [37], very recent studies increasingly explore causality-aware, robustness-oriented, and representation-level debiasing strategies. For example, causal graph diffusion models have been proposed to improve out-of-distribution recommendation by capturing structural dependencies beyond observational interactions [39]. Motivation-aware retrieval approaches further emphasize aligning long-tail exposure with users' latent intent rather than relying solely on interaction frequency [40]. Similarly, causal intervention frameworks address the evolving nature of person popularity, highlighting the temporal dynamics of exposure bias [38].

In parallel, unbiased learning-to-rank methods based on conditional mutual information minimization seek to disentangle relevance from spurious popularity correlations [41], while adversarial graph-dropout techniques improve robustness in graph-based collaborative filtering [42].

Recent work also demonstrates that popularity bias persists even in large language model-based recommenders, suggesting that exposure imbalance remains a fundamental challenge across emerging recommendation paradigms [43]. Temporal recommenders, including the Netflix Prize-winning approaches, demonstrate that user preferences evolve over time; however, these models primarily optimize predictive accuracy rather than user-level exposure calibration, leaving popularity alignment as a complementary challenge.

Collectively, these studies, including temporal, causal, and robustness-oriented approaches, indicate a broader shift toward integrating personalization-aware mechanisms into popularity-debiasing pipelines. SUPER aligns with this emerging direction by emphasizing user-level exposure calibration while maintaining model-agnostic applicability under standard offline evaluation settings.

To address this gap, we propose *SUPER*, which combines Pareto-based user-centric popularity profiling with a hybrid pre- and post-processing framework. This design enables fine-grained control over head–tail exposure at the individual level while remaining compatible with multiple backbone recommenders and standard offline evaluation protocols. Rather than competing within a single methodological paradigm, *SUPER* is designed to complement existing debiasing strategies by introducing a lightweight user-centric calibration layer that operates independently of the underlying recommender. In this sense, the framework can be interpreted as a practical bridge between classical model-agnostic exposure control and emerging user-aware optimization

approaches, enabling stable popularity alignment without imposing strong causal assumptions or architectural constraints.

Taken together, these developments suggest a shift from heuristic exposure balancing toward more structured and user-aware paradigms.

III. THE PROPOSED SMART USER-CENTRIC POPULARITY DEBIASING METHOD

One of the main objectives in tackling popularity bias is to ensure that recommendation lists reflect each user's observed preferences regarding item popularity. While many existing techniques simply adjust item rankings post-hoc or rely on global debiasing strategies, our approach provides a more personalized and principled solution. By integrating the Pareto principle with user-specific popularity inclinations, we create a smart recommendation framework that harmonizes both popular and less-known (tail) items within the final recommendations.

SUPER is not intended as a causal debiasing approach; rather, it is a calibration-oriented framework designed for standard offline recommendation settings where impression/exposure logs are unavailable. Accordingly, *SUPER* focuses on controllable exposure calibration under interaction-only benchmarks, where structurally separating head and tail interactions provides a practical way to reduce representation imbalance prior to ranking-time calibration.

Figure 1 provides an overview of the *SUPER* pipeline, illustrating the offline head–tail partitioning, user popularity inclination estimation, and the blueprint-based merging strategy used to construct the final recommendation list.

A. PRE-PROCESSING: SEPARATING THE DATASET INTO POPULAR AND TAIL ITEMS

We begin by applying the Pareto principle to partition the entire item catalog into two distinct subsets:

- 1) **Popular (Head) Items (H):** Items are sorted in descending order according to their rating frequency, and the “popular” set is defined as the minimal prefix of this sorted list whose cumulative rating count reaches 20% of all ratings in the dataset. This means that popular items are those accounting for 20% of the total interaction volume, rather than the top 20% of items by count, capturing the mainstream portion of the catalog that receives disproportionately high attention
- 2) **Tail Items (T):** The remaining 80% of the catalog, items that have not attained widespread consumption, form the long-tail set. These items, though less frequently interacted with, can be crucial for delivering diverse and personalized recommendations to users with unique tastes.

This preprocessing step clearly distinguishes between mainstream (popular) and niche (tail) content. By isolating the two sets upfront, we reduce the risk that the dominance of

popular items overshadows the learning of tail-item signals in models specialized for niche content.

Although the head–tail partitioning may appear structurally simple, it serves a principled role in mitigating representation imbalance during model learning. In popularity-skewed datasets, jointly training on mixed interactions often biases latent representations toward head items, suppressing informative signals from the long-tail. By separating these subsets, the proposed framework introduces an implicit exposure regularization effect that stabilizes learning dynamics prior to calibration. The objective of this design is therefore not increased architectural complexity, but controlled representation learning that preserves minority-item signals while remaining compatible with diverse backbone recommenders.

B. TRAINING ON POPULAR AND TAIL SUBSETS SEPARATELY

After dividing the dataset, we train distinct recommendation models, one using only the popular subset and another using only the tail subset. This approach allows each model to specialize and learn latent patterns more effectively without being biased by the other subset's distribution. For example:

- **Popular-Only Model:** Trained on the H set, it excels at recommending widely-liked items.
- **Tail-Only Model:** Trained on the T set, it focuses on uncovering hidden gems and niche content that might otherwise be overshadowed by popular items.

Training these models independently decouples the learning processes, ensuring that the tail-specialized model can gain a robust understanding of lesser-known content without the interference of highly rated mainstream items. This decoupled training is intended to prevent popularity-dominant gradients from leaking into tail representations, which can occur under a shared embedding space. In this sense, the split-and-train design functions as a structural constraint that improves controllability of head–tail exposure before the final user-level calibration step.

Algorithm 1 summarizes this pre-processing stage by (i) applying Pareto-based head–tail partitioning and (ii) training separate models on the popular and tail subsets.

We note that the popular-item and tail-item components in *SUPER* are intentionally trained independently. While cross-subset knowledge transfer or shared representations could potentially improve raw tail-item accuracy, such mechanisms may also reintroduce popularity signals into tail representations and thereby weaken exposure control. Following common practice in popularity-bias mitigation, *SUPER* prioritizes controlled popularity exposure through decoupled modeling and post-processing calibration, rather than joint representation learning. Exploring cross-subset transfer mechanisms is therefore treated as a complementary research direction and left for future work.

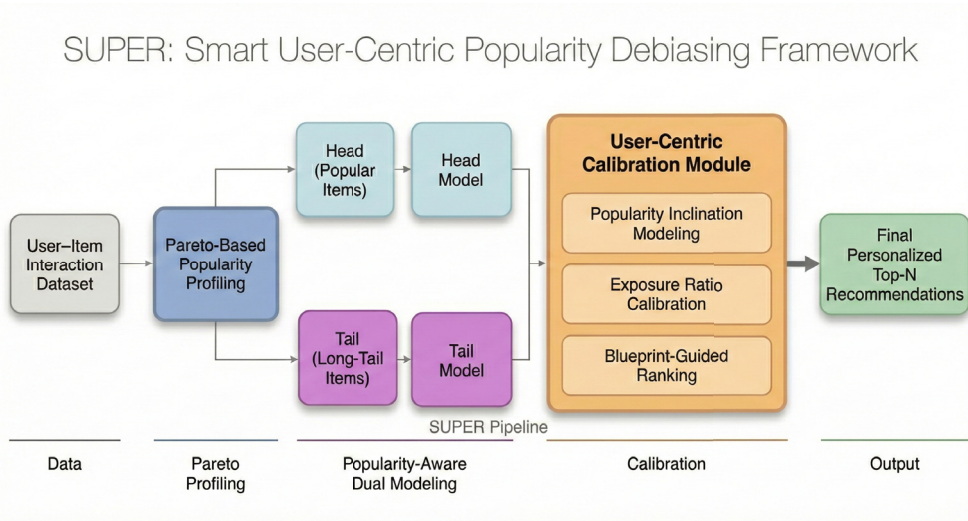


FIGURE 1. Overview of the proposed SUPER hybrid framework for popularity debiasing. The pipeline integrates Pareto-based popularity profiling, specialized dual modeling, and a user-centric exposure calibration module to construct personalized Top-N recommendations while maintaining alignment with observed user popularity tendencies.

1) COMPLEXITY ANALYSIS (ALGORITHM 1)

Let n denote the number of items in the catalog (or the number of candidate items scored per user at inference time), m the number of observed interactions (ratings) in the dataset, f the latent dimension of the backbone model, N the target recommendation list size (e.g., $N=10$), and M the size of an optional intermediate candidate set (if used). We report complexities separately for (i) offline preprocessing/training steps that scale with (n, m, f) and (ii) online per-user list construction that scales primarily with n (scoring/sorting) and N (merging).

- Sorting items by popularity: $O(n \log n)$, where n is the number of items.
- Extracting subsets D_{pop} , D_{tail} : Approximately $O(m)$, where m is the total number of interactions.
- Training models (backbone-dependent): The training cost is algorithm-specific. For example, matrix-factorization-based models are often expressed as $O(mf)$ per epoch, where m denotes the number of interactions and f the latent dimension, while neural or graph-based recommenders may exhibit different scaling behavior. Accordingly, we report training complexity as a model-dependent estimate rather than a strict theoretical bound.

The above complexities are provided as coarse, illustrative approximations intended to communicate relative computational trends. Exact costs may vary depending on the backbone model, optimization procedure, number of training epochs, and implementation details.

This phase is offline and does not need to be repeated for each user.

Importantly, *SUPER* does not introduce additional iterative training loops beyond the backbone recommender. Its extra computational cost is limited to lightweight head–tail

partitioning and rank-level merging, meaning that overall runtime is typically dominated by backbone training and per-user ranking operations.

C. ESTIMATING USER POPULARITY INCLINATION

A key insight of our approach is that not all users value popularity equally. While some may prefer mainstream hits, others might favor more specialized or under-the-radar offerings. To capture these differences, we estimate each user’s popularity inclination using the Pareto-based classification of items.

For each user u , we define the set of *consumed items* as the items with an observed interaction (e.g., an explicit rating) in the offline dataset. Formally,

$$C_u = \{i \mid r_{u,i} \text{ is observed}\}.$$

Since impression logs are not available in the public benchmarks used in this study, C_u provides a behavioral proxy for popularity inclination. The user’s popularity-inclination ratio is then computed as

$$Pop_u = \frac{|C_u \cap H|}{|C_u|},$$

where H denotes the set of popular (head) items defined in Algorithm 1.

For example, if a user’s interaction history contains 30% head items and 70% tail items, then $Pop_u = 0.3$. This ratio serves as a personalized calibration signal indicating how strongly the user’s observed behavior leans toward popular versus tail content, and it guides the subsequent head–tail blending during top- N construction. Importantly, to reduce preference–exposure conflation in evaluation, our popularity-calibration metric RMSE-PC is computed on *enjoyed items* (items rated above the user’s mean), as described in Section IV.

Identifiability Consideration: We emphasize that recovering a user's causal preference for popularity from observational interaction logs is fundamentally challenging in the absence of impression or exposure data. Historical interactions may reflect not only genuine interest but also visibility effects, position bias, and popularity reinforcement.

Accordingly, *SUPER* does not interpret the estimated popularity inclination as a ground-truth latent preference. Instead, it is treated as a behavioral proxy that serves as an operational calibration signal for controlling head–tail exposure at the user level. This perspective aligns with the standard offline evaluation setting commonly adopted in popularity-bias research, where behavioral interaction patterns are used as practical estimators of user tendencies when exposure logs are unavailable.

D. CONSTRUCTING THE FINAL TOP-N RECOMMENDATIONS

With the user-specific inclination Pop_u in hand, we produce the final top-N recommendation list by proportionally blending items predicted by the popular-only and tail-only models. The code provided in the integration phase performs the following steps:

- 1) **Generate Predictions Separately:** Using the previously trained models, we produce two sets of rating predictions for each user:
 - **Popular Predictions:** From the popular-only model.
 - **Tail Predictions:** From the tail-only model.
- 2) **Determine User's Proportion for Popular and Tail Items:** Given the user's popularity inclination Pop_u , we translate this ratio into a distribution for the final top-N list. For a top-N size (e.g., $N=10$), if $Pop_u = 0.3$, we select $N \times 0.3 = 3$ items from the popular predictions and $N \times 0.7 = 7$ items from the tail predictions.
- 3) **Selecting the Top Items:** Using the post-processing approach in Algorithm 2, we first compute $N_{pop} = \lfloor N \times Pop_u \rfloor$ and $N_{tail} = N - N_{pop}$. We then rank the items predicted by $\mathcal{M}_{pop}$ in descending order and choose the top N_{pop} . Similarly, we rank the items predicted by $\mathcal{M}_{tail}$ in descending order and select the top N_{tail} . By adhering to these user-specific proportions, we ensure that each final recommendation list reflects both the user's inclination toward mainstream versus niche content, and the system's objective of blending popular and tail items in a balanced manner.
- 4) **Combine and Output the Final Recommendations:** Rather than simply concatenating popular and tail selections in a static proportion, we follow a *blueprint-based merge* that respects how the user historically alternates between head and tail items. Specifically, we retrieve the user's real rating list L_u , sorted by descending preference, and note the positions at which

Algorithm 1 *SUPER* Pre-Processing

Require: Dataset $\mathcal{D}$ (columns {user_id, item_id, ...}), Pareto threshold α (e.g., 0.20)

Ensure: Partitioned sets ($\mathcal{D}_{pop}, \mathcal{D}_{tail}$) and trained models ($\mathcal{M}_{pop}, \mathcal{M}_{tail}$)

- 1: threshold $\leftarrow \mathcal{D}.shape[0] \times \alpha$ ▷ Compute interaction-volume threshold for head vs. tail
- 2: counts $\leftarrow$ count item_id frequencies in $\mathcal{D}$ ▷ Number of user interactions per item
- 3: sorted_counts $\leftarrow$ sort(Counts, Desc) ▷ Rank items from most to least popular
- 4: popular_items, long-tail_items $\leftarrow \emptyset$
- 5: total_ratings $\leftarrow 0$
- 6: **for** (item_id, count) $\in$ sorted_counts **do**
- 7: total_ratings $\leftarrow$ total_ratings + count
- 8: **if** total_ratings < threshold **then**
- 9: popular_items.append(item_id)
- 10: **else**
- 11: long-tail_items.append(item_id)
- 12: **end if**
- 13: **end for**
- 14: $\mathcal{D}_{pop} \leftarrow$ Filter($\mathcal{D}$, popular_items) ▷ All records for items deemed popular
- 15: $\mathcal{D}_{tail} \leftarrow$ Filter($\mathcal{D}$, long-tail_items) ▷ All records for items deemed less popular
- 16: $\mathcal{M}_{pop} \leftarrow$ Train($\mathcal{D}_{pop}$) ▷ Model specialized for head items
- 17: $\mathcal{M}_{tail} \leftarrow$ Train($\mathcal{D}_{tail}$) ▷ Model specialized for tail items
- return** ($\mathcal{D}_{pop}, \mathcal{D}_{tail}, \mathcal{M}_{pop}, \mathcal{M}_{tail}$)

Note: The Pareto threshold is defined over cumulative interaction volume rather than item count, ensuring that the head set captures the items responsible for the majority of observed user activity.

the user has historically consumed popular (*head*) items versus those for which the user has opted for niche (*tail*) options. Here, L_u is constructed exclusively from the user's observed historical interactions in the training data and contains only items that the user has actually rated or interacted with; it never includes unrated or test-only items. For each position in the top-N list, we then pull the highest-scoring candidate from the *popular* or *tail* pool based on whether that position is designated as *head* or *tail* in the user's original consumption pattern. This step ensures that each user's final top-N recommendations align with their personal ratio of head-to-tail items, reflecting the user's observed behavioral transitions between popular and niche content rather than enforcing a uniform, one-size-fits-all allocation. As a result, the system delivers a more accurate and user-centric experience, reflecting individual popularity inclination while maintaining diverse coverage. Crucially,

Algorithm 2 *SUPER* Post-Processing (Hard Quota, Soft Blueprint)

Require: User popularity inclination Pop_u , Popular predictions $\mathcal{P}_{pop}(u)$ (descending), Tail predictions $\mathcal{P}_{tail}(u)$ (descending), Real rating list L_u (descending), Target list size N , Popular item set H

Ensure: Final top- N recommendation list $RecList(u)$

```

1:  $N_{pop} \leftarrow \lfloor N \cdot Pop_u \rfloor, N_{tail} \leftarrow N - N_{pop}$ 
2:  $\mathcal{C}_{pop} \leftarrow \mathcal{P}_{pop}(u)[1:N_{pop}], \mathcal{C}_{tail} \leftarrow \mathcal{P}_{tail}(u)[1:N_{tail}]$ 
3:  $idx_{pop} \leftarrow 1, idx_{tail} \leftarrow 1, RecList(u) \leftarrow \emptyset$ 
4:  $B \leftarrow \min(N, |L_u|)$ 
5: for  $k \leftarrow 1$  to  $B$  do
6:    $b_k \leftarrow \mathbb{I}[L_u[k] \in H]$   $\triangleright$  blueprint: head(1)/tail(0)
7:   if  $b_k = 1$  and  $idx_{pop} \leq N_{pop}$  then
8:      $\text{append } \mathcal{C}_{pop}[idx_{pop}]; idx_{pop} \leftarrow idx_{pop} + 1$ 
9:   else if  $b_k = 0$  and  $idx_{tail} \leq N_{tail}$  then
10:     $\text{append } \mathcal{C}_{tail}[idx_{tail}]; idx_{tail} \leftarrow idx_{tail} + 1$ 
11:   else if  $idx_{pop} \leq N_{pop}$  then
12:     $\text{append } \mathcal{C}_{pop}[idx_{pop}]; idx_{pop} \leftarrow idx_{pop} + 1$ 
13:   else
14:     $\text{append } \mathcal{C}_{tail}[idx_{tail}]; idx_{tail} \leftarrow idx_{tail} + 1$ 
15:   end if
16: end for
17: while  $|RecList(u)| < N$  and  $idx_{pop} \leq N_{pop}$  do
18:    $\text{append } \mathcal{C}_{pop}[idx_{pop}]; idx_{pop} \leftarrow idx_{pop} + 1$ 
19: end while
20: while  $|RecList(u)| < N$  and  $idx_{tail} \leq N_{tail}$  do
21:    $\text{append } \mathcal{C}_{tail}[idx_{tail}]; idx_{tail} \leftarrow idx_{tail} + 1$ 
22: end while
23: return  $RecList(u)$ 

```

Note: The final list satisfies a hard quota / soft blueprint rule. The quotas (N_{pop}, N_{tail}) derived from Pop_u are enforced strictly, while the blueprint extracted from the first $B = \min(N, |L_u|)$ ranks of L_u is used only as a rank-level preference for choosing the next pool as long as the corresponding quota remains. If the blueprint requests a pool whose quota is exhausted, the algorithm deterministically switches to the other pool to maintain validity while preserving the hard composition.

the overall head–tail composition is enforced by *hard user-specific quotas* derived from Pop_u , while the blueprint serves only as a *soft rank-level ordering prior*. Under the leave-one-out protocol, the blueprint list L_u is constructed strictly from the *training* interactions of user u (i.e., excluding the held-out test item), ensuring that the merging prior does not access evaluation labels.

The complete post-processing and blueprint-based merging procedure described above is formally summarized in Algorithm 2.

1) BLUEPRINT-BASED MERGING: INTUITION AND EXAMPLE

Algorithm 2 constructs the final recommendation list using a *hard quota / soft blueprint* strategy. First, the user-specific

popularity inclination Pop_u determines a *hard* head–tail composition for the final top- N list via $N_{pop} = \lfloor N \cdot Pop_u \rfloor$ and $N_{tail} = N - N_{pop}$. This quota is enforced strictly: even if the user’s historical list L_u is dominated by head items at the top ranks, once the head quota is satisfied, the remaining positions must be filled from the tail pool (and vice versa).

Given these hard quotas, the *blueprint* derived from L_u is used only as a *soft, rank-level ordering prior* that decides *which pool to draw from next* as long as the corresponding quota is still available. Concretely, the algorithm first truncates the two candidate pools to $\mathcal{C}_{pop}$ and $\mathcal{C}_{tail}$ containing the top- N_{pop} and top- N_{tail} items from the popular-only and tail-only models, respectively. It then scans the first $B = \min(N, |L_u|)$ positions of L_u to obtain a head/tail indicator at each rank, and attempts to follow this indicator when selecting the next item. If the blueprint requests a pool whose quota is already exhausted, the method deterministically falls back to the other pool, thereby maintaining a valid top- N list while preserving the hard head–tail composition dictated by Pop_u .

From a conceptual perspective, this blueprint acts as a lightweight *positional regularizer*: proportional allocation alone controls the aggregate head–tail ratio but does not constrain *where* exposure occurs across ranks, which may lead to position-level miscalibration. By introducing rank-aware guidance without altering backbone relevance scores, the soft blueprint stabilizes exposure distribution across positions, while the hard quota guarantees that the final list always matches the user-specific popularity inclination. For short or noisy histories, we apply the blueprint only for the first $\min(N, |L_u|)$ ranks and fill any remaining slots using the residual user-specific quotas. If the blueprint covers fewer than N positions (i.e., $|L_u| < N$), the remaining ranks are filled greedily from the residual pools while strictly satisfying the hard quotas (N_{pop}, N_{tail}); in this completion phase, the exact within-quota ordering is not constrained by the blueprint.

2) WORKED EXAMPLE

Consider a target list size $N = 6$ and a user with $Pop_u = 0.5$, yielding $N_{pop} = 3$ and $N_{tail} = 3$. Assume the user’s real rating list L_u induces the following head–tail blueprint over ranks:

$$L_u = [\text{Pop}, \text{Tail}, \text{Pop}, \text{Pop}, \text{Tail}, \text{Tail}].$$

Let the quota-limited candidate pools (sorted by predicted relevance within each pool) be

$$\mathcal{C}_{pop}(u) = [p_1, p_2, p_3], \quad \mathcal{C}_{tail}(u) = [t_1, t_2, t_3].$$

The list is then constructed under a *hard quota / soft blueprint* rule: at each rank, the blueprint requests either head or tail, but the algorithm selects from that pool *only if* its remaining quota allows; otherwise it deterministically switches to the other pool. For the above blueprint, all requests are feasible within the quotas, hence scanning L_u yields

$$RecList(u) = [p_1, t_1, p_2, p_3, t_2, t_3].$$

Importantly, if the blueprint were to request more than N_{pop} head items within the first N positions, only the first N_{pop} selections could be drawn from $C_{pop}(u)$; once the head quota is met, the remaining positions must be filled from $C_{tail}(u)$ to satisfy the user-specific composition implied by Pop_u .

3) COMPLEXITY ANALYSIS (ALGORITHM 2):

For each user, the backbone(s) produce predicted scores over candidate items. In the worst case, scoring and sorting over all items incurs $O(n)$ and $O(n \log n)$ time, respectively. SUPER itself does not add an additional global optimization step; after obtaining the per-pool ranked lists, it truncates them to C_{pop} and C_{tail} of sizes N_{pop} and N_{tail} with $N_{pop} + N_{tail} = N$, and performs the hard-quota / soft-blueprint merge in $O(N)$ time. Therefore, the additional overhead of SUPER beyond backbone scoring is $O(N)$ per user, and overall runtime is dominated by backbone inference and (when applicable) large-scale sorting (Table 3).

E. BENEFITS OF THE SMART USER-CENTRIC APPROACH

This method addresses several limitations of prior techniques:

User-Centric Calibration: By determining how many recommendations should come from the popular set versus the tail set for each individual user, we ensure that recommendations align more closely with that user's observed consumption distribution.

Fair Representation of the Long-Tail: Our approach naturally incorporates tail items into the recommendation process, promoting diversity and providing opportunities for lesser-known items to surface.

Mitigating Popularity Bias: Rather than merely down-ranking popular items or imposing blanket constraints, this solution trains models separately on both subsets and then tailors the blend of final recommendations. The result is a more nuanced debiasing that accounts for each user's own level of comfort with mainstream or niche items.

Scalable and Adaptable: The approach is flexible and can be applied to various datasets, domains, and underlying recommendation algorithms. The pre-processing does not assume a specific model architecture, making the framework portable and easily integrated into different scenarios.

More broadly, SUPER can be viewed as a user-adaptive exposure control framework that operates at the ranking layer. Instead of relying solely on global exposure constraints or stochastic reallocation, the method introduces a structured yet lightweight regularization signal that preserves alignment between observed behavioral patterns and recommendation exposure. This perspective positions SUPER within a growing class of calibration-oriented recommender approaches that emphasize exposure coherence alongside ranking accuracy.

F. ILLUSTRATIVE SCENARIO

To clarify how the SUPER framework operates in practice, consider a hypothetical user whose historical records indicate a moderate preference for popular (head) items relative to

niche (tail) ones. Suppose, for instance, that this user has interacted with 200 items in total, with 130 of those classified as popular and the remaining 70 belonging to the tail. From these statistics, one infers that the user's *popularity inclination ratio* is

$$Pop_u = \frac{130}{200} = 0.65.$$

Hence, approximately 65% of this individual's consumed items are from the head portion of the catalog, while 35% are from the tail.

1) PRE-PROCESSING AND PREDICTION

Following the SUPER design, two specialized models are trained: one ($\mathcal{M}_{pop}$) exclusively on the head-dominant subset and another ($\mathcal{M}_{tail}$) exclusively on the tail-dominant subset. At recommendation time, each model produces a ranked list of items for the user (for instance, top-10 suggestions per model):

- *Popular-Only Model Output:* A descending list of items predominantly aligned with the head part of the catalog.
- *Tail-Only Model Output:* A descending list emphasizing less popular or niche items.

2) DETERMINING USER-SPECIFIC PROPORTIONS

Since $Pop_u = 0.65$, if the system is to produce a top-10 recommendation list, it will compute

$$N_{pop} = \lfloor 10 \times 0.65 \rfloor = 6, \quad N_{tail} = 10 - 6 = 4.$$

This calculation instructs SUPER to select 6 items from the user-specific popular-only ranking and 4 items from the tail-only ranking.

3) CONSTRUCTING THE SMART RECOMMENDATION LIST

While each model's output ($\mathcal{M}_{pop}$ and $\mathcal{M}_{tail}$) is first sorted in descending order of predicted relevance, the final merge does not merely concatenate top- N_{pop} and top- N_{tail} . Instead, we reference the user's historical *blueprint* of head-tail consumption to guide how these items are interleaved. Suppose, for example, that positions in the user's real rating list alternately reflect head or tail preferences. Then for each rank k in the top- N recommendation, we:

- 1) Determine whether the k^{th} historical position is head or tail in the user's consumption blueprint.
- 2) If it is head, pick the next highest-scoring item from $\mathcal{M}_{pop}$'s descending list.
- 3) If it is tail, choose the next highest-scoring item from $\mathcal{M}_{tail}$'s descending list.

This process yields a smart top- N list that matches the user's *empirically observed ratio* of interest in head vs. tail content without enforcing a uniform or arbitrary blend. By anchoring the final merge to the user's documented head-tail trajectory, the system more faithfully reflects individualized consumption patterns, thereby ensuring that diverse, personalized recommendations emerge in a manner consistent with the user's actual historical preferences.

TABLE 1. Combined complexity of key components.

Step	Description	Complexity
1	Offline Pareto classification into H and T sets	$O(n \log n)$
2	Constructing D_{pop} and D_{tail}	$O(m)$
3	Training models (popular-only and tail-only)	Model-dependent (backbone-specific $\sim O(mf)$)
4	Generating prediction scores (for each user)	Model-dependent (often $O(n)$)
5	Sorting prediction scores to select top M candidate items	$O(n \log n)$
6	Computing user's popularity ratio and partitioning final N	$O(1)$
7	Selecting top- N from popular and tail lists and combining them	$O(N)$

4) OUTCOME AND RATIONALE

By adhering to the user's estimated *popularity inclination ratio*, *SUPER* tailors the final recommendation to reflect each person's established preferences while introducing an appropriate degree of diversity. This calibration-based mechanism simultaneously combats the tendency to overemphasize prevalent items (thereby mitigating popularity bias) and ensures that the user's legitimate tastes inform the ultimate combination of head and tail items. In so doing, the user is more likely to encounter content that resonates with their observed tastes, encouraging an enriched and balanced discovery experience across the broader catalog.

G. COMBINED COMPLEXITY TABLE

The Table 1 summarizes the complexity of key components from the two-phase approach.

Notation for Table 1: Here, n denotes the number of items in the catalog (or scored candidates per user), m the total number of interactions, f the latent dimension of the backbone model, N the final recommendation length (Top- N), and M an optional intermediate candidate size (e.g., if a method first selects top- M items before constructing the final Top- N list).

Given the dataset sizes in Table 2, the dominant cost for *SUPER* is the per-user ranking step over candidate items followed by sorting, which scales primarily with the item catalog size and the number of users. Thus, datasets with larger item sets (e.g., DoubanBooks) naturally incur higher wall-clock time in top- N generation, while the training cost remains identical to the underlying backbone recommender.

After completing the offline training and classification steps, the online process involves generating user-specific top- N recommendations aligned with their personal popularity inclination. By doing so, the proposed Smart User-Centric approach offers a more balanced, diverse, and user-aligned solution to the popularity bias problem.

IV. EXPERIMENTAL SETUP

In this section, we present the experimental setup designed to assess the effectiveness of our *Smart User-centric Popularity Exposure Reduction for Fair and Diverse Recommendations*. We structure this evaluation by detailing the datasets employed, the chosen evaluation metrics, the experimentation methodology, and the baseline methods against which we compare our proposed solution. The aim is to provide a comprehensive and fair assessment of the approach's ability

to mitigate popularity bias while remaining closely aligned with individual user preferences. In forming our experimental design, we draw inspiration from prior works that have tackled popularity bias and related challenges [10], [20], [27].

A. DATASETS

To ensure the generalizability of our findings, we rely on three widely studied benchmark datasets: MovieLens-1M (ML),¹ Douban Book (DB),² and Yelp.³ These datasets are commonly used in recommender system research and differ substantially in domain, size, and sparsity, thereby allowing us to examine how varying data characteristics influence the performance and robustness of our approach.

- **ML:** Focuses on movie preferences, featuring over one million user ratings on a five-star scale. Its moderately dense structure offers a stable testbed for evaluating the debiasing capabilities of our method.
- **DB:** Extracted from a large-scale online book community, this dataset includes user-item interactions also recorded on a five-star scale. Its higher sparsity and larger catalog size enable analysis under more challenging conditions.
- **Yelp:** Captures user reviews on local businesses, ranging from restaurants to service providers. The high sparsity and niche-oriented nature of these interactions make the Yelp dataset particularly suited to testing how well our approach handles long-tail and diverse user interests.

All rating values across these datasets are discrete and span the [1–5] interval. The diversity in size, average rating patterns, and sparsity ratios mirrors real-world scenarios, ensuring that our conclusions are not narrowly tied to a single domain [9], [21].

B. EVALUATION METRICS

We evaluate the proposed method using a compact yet comprehensive set of metrics that capture recommendation accuracy, popularity calibration, and beyond-accuracy objectives such as diversity and novelty. Following prior work on popularity bias and user-centric evaluation [8], [27], we deliberately omit Precision, Recall, and F1-score, as these metrics do not directly reflect popularity exposure or calibration effects.

¹<https://grouplens.org/datasets/movielens/1m/>

²<https://www.douban.com/>

³<https://www.yelp.com/>

TABLE 2. Detailed properties of the ML, DB, and Yelp datasets, including head–tail popularity statistics.

Metric	ML	DB	Yelp
# of users	6,040	13,024	16,239
# of items	3,952	22,347	14,284
# of ratings	1,000,209	792,062	198,397
Sparsity (%)	95.8	99.7	99.9
Avg. ratings/user	165.6	60.8	12.2
Avg. ratings/item	253.1	35.4	13.9
Head–Tail Split Statistics			
# of head items	113	221	185
# of tail items	3,839	22,126	14,099
Avg. ratings/head item	1771.6	718.4	215.0
Avg. ratings/tail item	208.4	28.6	11.3
Head interaction share	19.89	20	19.99
Head rating (min/med/max)	1269 / 1627 / 3428	392 / 573 / 2679	131 / 179 / 931
Tail rating (min/med/max)	1 / 114 / 1263	6 / 14 / 389	1 / 4 / 131

- **Root Mean Square Error in Popularity Calibration (Rmse-PC)** : [44] This metric measures how closely the popularity level of the recommended top- N items aligns with each user’s interaction-derived popularity-inclination proxy. Unlike interaction-based measures, Rmse-PC relies only on *enjoyed items* (items rated above the user’s mean rating), thereby reducing exposure-driven noise. Lower values indicate better popularity calibration.
- **Mean Rank Miscalibration (MRMC)**: MRMC evaluates how well the cumulative popularity distribution of a ranked recommendation list matches the user’s historical head–tail consumption pattern. It computes rank-wise divergence between historical and recommended popularity distributions and averages this miscalibration across users. Lower MRMC values correspond to better alignment.
- **nDCG (Normalized Discounted Cumulative Gain)**: We report nDCG to assess ranking quality in terms of relevance and position sensitivity [21], [33]. Higher values indicate better accuracy.
- **Average Percentage of Long-Tail Items (APLT)**: APLT measures the proportion of tail items in the top- N recommendation lists, reflecting the system’s tendency to promote niche content [45]. Higher values indicate stronger long-tail exposure.
- **Entropy**: Entropy quantifies the dispersion of recommended items across the catalog, based on the frequency with which items appear in users’ top- N lists [8]. Higher entropy indicates more balanced item exposure.
- **Long-Tail Coverage (LTC)**: LTC measures the fraction of tail items that appear at least once in the union of all recommendation lists [45], capturing global catalog-level coverage.
- **Novelty**: Novelty is computed as the average self-information of recommended items:

$$\text{Novelty}@N(u) = \frac{1}{N} \sum_{i \in \text{Rec}@N(u)} -\log_2 p(i), \quad (1)$$

where $p(i) = \frac{n_i}{\sum_{j \in I} n_j}$ and n_i is the number of training interactions for item i (computed on the training split only). Higher values indicate more novel recommendations.

• **General Key Performance Indicator (GKPI): A Comprehensive Performance Measure**

Recommender system performance is inherently multi-dimensional, as accuracy-oriented and beyond-accuracy metrics often reflect competing objectives. To facilitate a concise comparison across methods, we report the General Key Performance Indicator (GKPI) as a compact *summary measure* that aggregates multiple evaluation dimensions into a single score.

GKPI combines a core accuracy metric (nDCG) with four beyond-accuracy metrics (APLT, Novelty, Entropy, and LTC). Rather than replacing these metrics, GKPI serves as an auxiliary indicator that highlights whether a method achieves a balanced trade-off between relevance and popularity-aware objectives.

To avoid dominance by any single dimension, each beyond-accuracy metric is paired with nDCG using a harmonic-type mean:

$$\mathcal{H}(x, y) = \frac{2xy}{x + y}. \quad (2)$$

The harmonic mean penalizes extreme imbalance, ensuring that strong performance in one dimension cannot fully compensate for weak performance in another. This property is particularly suitable for popularity debiasing, where over-optimizing diversity or novelty may significantly harm relevance.

GKPI is then computed as the arithmetic mean of the four harmonic combinations between nDCG and each beyond-accuracy metric. Equal weighting is intentionally adopted to avoid introducing additional hyperparameters or domain-specific assumptions, and to maintain neutrality across evaluation dimensions in the absence of a universally accepted prioritization scheme.

Clarification on the Role of GKPI: We emphasize that GKPI is *not* an optimization objective, nor a replacement for individual metrics. All substantive conclusions in this study

are derived from standard evaluation measures, including $nDCG$, RMSE-PC, MRMC, APLT, Entropy, LTC, and Novelty. GKPI is reported solely as a high-level summary to support interpretability when multiple metrics exhibit trade-offs. In this sense, GKPI should be interpreted as a *comparative rubric* for overall balance rather than a ground-truth measure of user satisfaction. It is particularly useful when methods exhibit conflicting metric-level behavior (e.g., higher novelty but lower $nDCG$), providing a single summary that rewards *simultaneous* strength across relevance and beyond-accuracy dimensions.

All of the discussed beyond-accuracy metrics, APLT, Entropy, Novelty, and LTC, benefit from higher values, generally signifying a more diverse, less biased, and more dynamic recommendation environment. By employing these measures in conjunction, one gains a comprehensive perspective on how effectively the system tackles popularity bias while simultaneously catering to varied user preferences and stimulating content exploration. Meanwhile, the calibration metrics *Rmse-PC* and *MRMC* gauge how closely recommended items' popularity aligns with each user's underlying inclinations. Taken together, beyond-accuracy and calibration indicators provide a well-rounded assessment of *trade-offs* among recommendation goals (relevance, exposure calibration, and diversity-related objectives) under standard offline evaluation.

Moreover, the *GKPI* integrates these multiple dimensions, accuracy ($nDCG$) and beyond-accuracy measures (APLT, Entropy, Novelty, LTC), into a single, consolidated index, offering a streamlined yet holistic view of system performance. By balancing user-centric popularity calibration with broader objectives like diversity and novelty, the *GKPI* facilitates direct comparisons among different debiasing strategies, helping to identify approaches that deliver both rich user experiences and equitable content exposure.

C. EXPERIMENTATION METHODOLOGY

Our experimental design follows standard practices in recommendation system evaluation while incorporating specific considerations for popularity bias and user-centricity. We employ a leave-one-out evaluation scheme [46], which is widely regarded as a robust protocol in the literature. For each user, we set aside one interaction for testing and use the remaining interactions for training. This ensures that the evaluation is both user-specific and scenario-agnostic, allowing for fine-grained insights into how each approach performs across varied user profiles.

For all backbone recommenders, hyperparameters were selected using literature-recommended search ranges and a held-out validation split. The same tuning protocol was applied consistently across all datasets. Importantly, all popularity-debiasing methods were evaluated on top of identical tuned base recommenders, ensuring that performance differences reflect the debiasing strategies rather than disparities in model optimization. Since *SUPER* operates as a

post-processing method, it introduces no additional trainable parameters beyond those of the base recommender.

Although *SUPER* does not introduce any additional learning objectives, loss functions, or optimization procedures beyond those of the backbone recommenders, it requires training the same backbone model separately on the head and tail item subsets. These models share identical architectures and hyperparameters, and differ only in the training data partition. The popularity-aware behavior of *SUPER* is achieved exclusively through a lightweight post-processing merge, without modifying the underlying model structures or training objectives.

Following the training phase, we use the models to predict user preferences for all candidate items. We then directly construct the top- N (where typically $N = 10$) recommendation list for each user by blending the outputs of the popular-only and tail-only models based on the user's derived popularity inclination ratio (as described in Section III). Unlike some existing methods that first select a large candidate set (e.g., top-100) before final refining [20], [47], our approach integrates popularity inclination into the recommendation generation process seamlessly, reducing complexity and ensuring closer alignment with user tastes from the outset.

By re-ranking items according to user-specific distributions between popular and tail content, the system can dynamically adjust each top- N list. This method does not rely on static thresholds or one-size-fits-all debiasing rules, but rather adapts to evolving user interests. The final recommendation lists are evaluated using the aforementioned metrics, yielding a comprehensive performance profile.

D. BASELINE METHODS AND COMPARED APPROACHES

To assess the effectiveness of the proposed *SUPER* framework, we compare it against a diverse set of baseline recommenders and state-of-the-art popularity-debiasing approaches. The selected baselines cover different modeling paradigms, including probabilistic factorization, neural models, clustering-based methods, and pairwise ranking approaches. This diversity ensures that the observed results are not specific to a particular recommender family.

1) BASELINE RECOMMENDERS

We employ the following widely used recommender models as backbones:

- **HPF** [48]: A probabilistic matrix factorization model designed for implicit feedback and sparse data.
- **SKMeans** [49]: A clustering-based recommender that groups users or items using cosine similarity.
- **VaeCF** [50]: A variational autoencoder-based collaborative filtering model capturing non-linear user-item interactions.
- **NeuMF** [51]: A neural hybrid model combining generalized matrix factorization and multi-layer perceptrons.

- **WBPR** [52]: A pairwise ranking approach that incorporates item-dependent weighting.

These models serve as representative baselines covering probabilistic, neural, clustering-based, and ranking-oriented recommendation strategies.

2) POPULARITY-DEBIASING BENCHMARKS

We compare SUPER against established post-processing popularity-debiasing methods that operate independently of the backbone recommender:

- **FA*IR** [35]: A fairness-aware re-ranking method that enforces balanced exposure between head and tail items.
- **Discrepancy Minimization (DM)** [34]: An exposure-balancing approach that aligns item recommendation frequencies with a target distribution.
- **xQuad** [20]: A diversification-based re-ranking strategy that balances relevance and popularity categories.
- **Aug** and **Mul** [47]: Weight-based re-ranking methods that penalize popular items using additive or multiplicative adjustments.
- **VaR** [4], [32]: A value-aware ranking approach favoring long-tail items through inverse popularity weighting.
- **CausE** [53]: A causal embedding method that disentangles preference from exposure bias.
- **DICE** [54]: A disentanglement-based model separating user interest from conformity effects.
- **MACR** [55]: A counterfactual post-processing approach correcting popularity-induced distortions.
- **PDRO** [36]: A distributionally robust optimization framework designed to stabilize recommendation quality under popularity-skewed interaction distributions.
- **PPAC** [37]: A personalized calibration approach that dynamically adjusts recommendation exposure according to user-specific popularity sensitivity.
- **CausalEPP** [38]: A causal debiasing method that penalizes exposure-driven interactions to better separate preference from popularity effects.

All baseline methods are evaluated under a unified experimental protocol with consistent parameter tuning based on their original publications. This ensures a fair and transparent comparison across different modeling and debiasing paradigms.

V. EXPERIMENTAL RESULTS

This section presents a comprehensive empirical evaluation of the proposed SUPER framework in the context of popularity bias mitigation. Our experimental design reflects the current state of the literature by incorporating a broad spectrum of debiasing strategies, ranging from classical re-ranking and exposure-balancing methods to recent user-aware, optimization-based, and causality-driven approaches.

Specifically, we compare SUPER against a diverse set of competing strategies covering multiple methodological paradigms, including re-ranking-based approaches (*AUG*, *MUL*, *VaR*, *DM*, *Fair*, *xQuad*), causality-

representation-driven methods (*CausE*, *DICE*, *MACR*), and recent user-aware optimization frameworks such as *PDRO*, *PPAC*, and *CausalEPP*.

All debiasing methods are evaluated on top of five widely adopted backbone recommenders, *HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR*, across three benchmark datasets: *ML* (MovieLens-1M), *DoubanBooks*, and *Yelp*. The diversity of both recommenders and datasets enables a robust examination of performance under varying sparsity levels, interaction densities, and domain characteristics.

All experiments are repeated with five random seeds to reduce sensitivity to stochastic effects, and we report seed-averaged mean performance for all methods. To assess whether SUPER's improvements are consistently observed across random initializations, we conduct paired Wilcoxon signed-rank tests over matched seeds. Each test therefore evaluates consistency rather than large-sample statistical power, yielding five paired observations per comparison.

For completeness, Wilcoxon tests are computed for all debiasing method pairs within each dataset-recommender-metric setting. To maintain clarity and avoid visual overload, we report only comparisons between SUPER and the *second-best* debiasing method, defined deterministically as the strongest non-SUPER baseline according to seed-averaged mean performance for the corresponding metric (higher-is-better for nDCG, APLT, Entropy, LTC, Novelty, and GKPI; lower-is-better for Rmse-PC and MRMC). This reporting strategy provides a conservative and focused assessment by benchmarking against the strongest competing alternative rather than weaker baselines. Statistical significance is annotated directly in the figures using star notation, where (*), (**), and (***) denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively, and is shown only when SUPER achieves the best debiasing performance for that setting.

Collectively, this experimental design prioritizes breadth of comparison, statistical robustness, and methodological fairness, thereby providing a reliable foundation for assessing whether user-centric, rank-aware popularity calibration can deliver consistent gains across diverse recommendation environments. Importantly, this design allows us to evaluate whether calibration-oriented exposure control can remain effective even when compared against strong, recently proposed debiasing frameworks.

Since popularity alignment is widely recognized as an important factor in recommendation quality, we begin by analyzing calibration performance before turning to beyond-accuracy and ranking metrics.

A. CALIBRATION RESULTS: Rmse-PC AND MRMC

Assessing how closely each debiasing approach tailors its recommendations to users' interaction-derived popularity tendency lies at the heart of this study. To evaluate this alignment, we consider two key metrics: *Rmse-PC* (Root Mean Square Error in Popularity Calibration) and *MRMC* (Mean Rank Miscalibration). While *Rmse-PC* measures how accurately the overall fraction of popular versus tail items

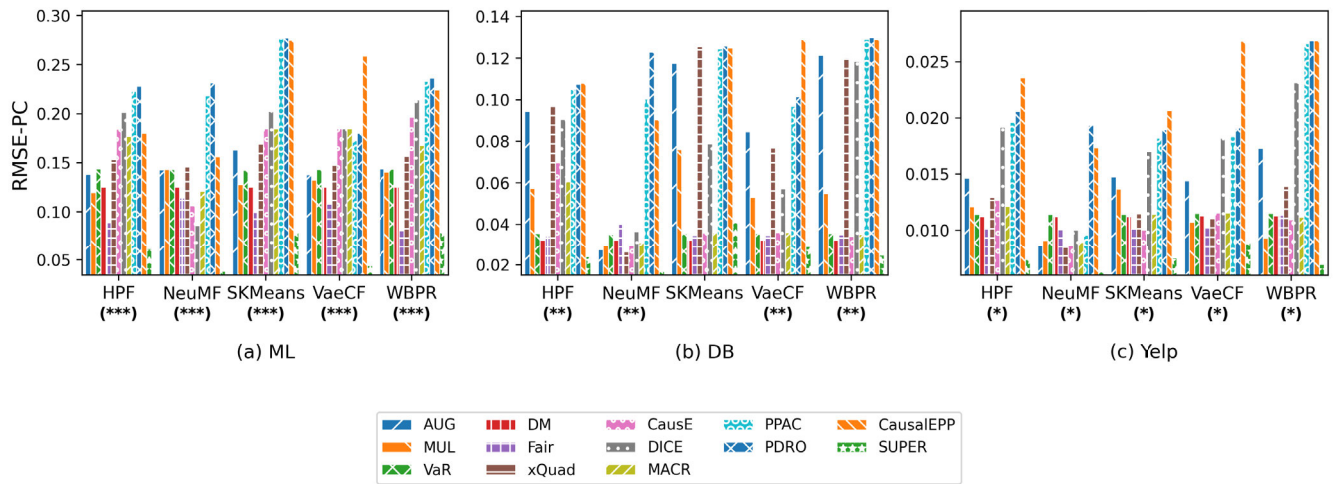


FIGURE 2. *Rmse-PC* results of the popularity-debiasing methods for the *HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR* algorithms on the (a) ML, (b) DB, and (c) Yelp datasets. Statistical significance is indicated using star notation, where (*), (**), and (***) denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively, based on paired Wilcoxon signed-rank tests across seeds; stars correspond to *SUPER* versus the second-best debiasing method for the corresponding setting. Stars are shown only when *SUPER* is the best-performing method in that setting; otherwise no star is displayed.

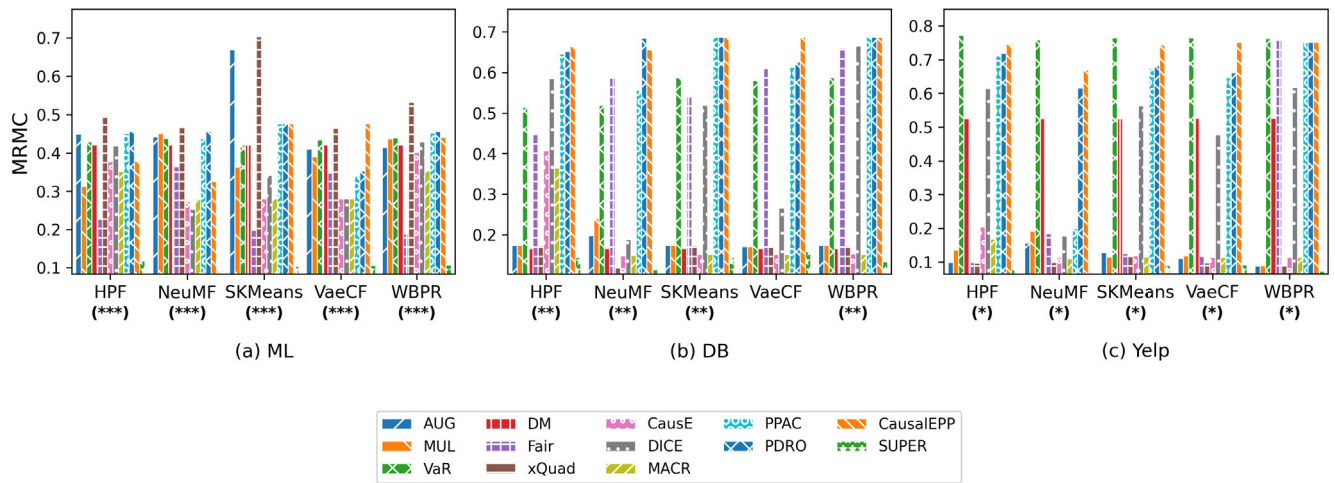


FIGURE 3. *MRMC* results of the popularity-debiasing methods for the *HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR* algorithms on the (a) ML, (b) DB, and (c) Yelp datasets. Statistical significance is indicated using star notation, where (*), (**), and (***) denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively, based on paired Wilcoxon signed-rank tests across seeds; stars correspond to *SUPER* versus the second-best debiasing method for the corresponding setting. Stars are shown only when *SUPER* is the best-performing method in that setting; otherwise no star is displayed.

matches a user's historical inclinations, *MRMC* examines whether that alignment holds consistently across the recommended ranks.

A global view of these metrics, presented in Figure 2 for *Rmse-PC* and Figure 3 for *MRMC*, indicates that *SUPER* achieves consistently lower calibration errors than a broad spectrum of popularity-debiasing strategies. Our experimental benchmark covers (i) re-ranking-based debiasing methods (*AUG*, *MUL*, *VaR*), (ii) representation- or causality-driven approaches (*CausE*, *DICE*, *MACR*), and (iii) user-aware optimization frameworks (*PDRO*, *PPAC*, *CausalEPP*).

Although *PDRO* and *PPAC* explicitly model robustness or personalized calibration objectives, their empirical behavior

reveals notable calibration instability. Across all datasets and backbone recommenders, these methods frequently exhibit substantially higher *Rmse-PC* values, indicating difficulties in aligning recommendation popularity with users' historical tendencies. The gap becomes even more pronounced under the rank-sensitive *MRMC* metric, where both methods often produce markedly elevated miscalibration scores. This pattern suggests that optimizing for distributional robustness or adaptive weighting alone may not sufficiently regulate where exposure occurs within the ranked list.

Similarly, while *CausalEPP* introduces a causal perspective by penalizing exposure-driven popularity signals, its calibration performance remains less stable than that of

SUPER, particularly in sparse settings where accurately disentangling preference from exposure becomes inherently challenging. These observations collectively indicate that stronger modeling assumptions do not necessarily translate into improved rank-level calibration.

In contrast, *SUPER* maintains consistently low calibration error by directly regulating head–tail exposure during the construction of each user’s top- N list. Rather than relying solely on global optimization objectives, the method combines user-specific popularity inclination with a rank-aware merging strategy, thereby preventing abrupt exposure shifts across positions.

A closer examination of Figure 2(a–b) reveals that on the *ML* and *DoubanBooks* datasets, *SUPER* outperforms nearly all competing approaches in terms of *Rmse-PC*. For instance, on the *ML* dataset with the *VaeCF* recommender, *SUPER* reduces *Rmse-PC* from 0.108 achieved by the second-best method (*Fair*) to 0.044, corresponding to an improvement of approximately 59%, while simultaneously yielding a lower *MRMC*. Such gains highlight the importance of position-aware exposure regulation, particularly in datasets characterized by heterogeneous user behavior.

This advantage becomes even more evident on the sparse *Yelp* dataset, where limited interaction histories amplify the risk of popularity misalignment. In this challenging regime, several competing approaches, including robustness-oriented and calibration-driven methods, display increased calibration errors. *SUPER*, by contrast, preserves stable *Rmse-PC* and *MRMC* values, suggesting that explicit rank-level control provides resilience against noisy behavioral signals.

Turning to Figure 3, we observe that *SUPER* consistently preserves calibration coherence throughout the ranked list. Methods such as *PDRO*, *PPAC*, *CausE*, *DICE*, and *MACR* may approximate the aggregate head-to-tail ratio, yet often exhibit rank-level volatility, with corrective adjustments concentrated in specific portions of the list. By contrast, *SUPER* achieves lower *MRMC*, indicating smoother and more behaviorally consistent exposure across ranks.

These findings suggest that controlling only the aggregate popularity proportion is insufficient; position-aware exposure regulation emerges as a critical requirement for achieving robust rank-level calibration. In this sense, *SUPER* can be interpreted as a *rank-aware popularity calibration* framework, where the user-level head–tail ratio constrains overall composition while the blueprint-based merge regularizes the positional structure of exposure.

Several factors contribute to this superior calibration behavior. First, *SUPER* treats user popularity inclination as a dynamic reference rather than a rigid global constraint, mitigating the risk of overcorrection commonly observed in optimization-heavy methods. Second, its effectiveness across diverse backbone recommenders (*HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR*) demonstrates that the calibration gains are not architecture-dependent. Finally, its stable performance in sparse domains highlights the advantage

of lightweight, user-aligned exposure control over more assumption-intensive strategies.

Overall, the calibration results demonstrate that *SUPER* more effectively captures the interplay between mainstream and niche interests at both the global and rank levels compared to re-ranking-based, causal, representation-driven, and robustness-oriented debiasing methods. By jointly minimizing *Rmse-PC* and *MRMC*, *SUPER* delivers recommendation lists that remain closely aligned with users’ interaction-derived popularity preferences throughout the entire top- N sequence.

B. BEYOND-ACCURACY DIMENSIONS: APLT, ENTROPY, LTC, AND NOVELTY

While effective calibration is vital for user satisfaction, recommender systems also benefit from promoting diversity, facilitating item discovery, and broadening coverage of less popular content. We therefore assess *APLT* (Average Percentage of Long-Tail Items), *Entropy*, *LTC* (Long-Tail Coverage), and *Novelty* to capture multiple beyond-accuracy dimensions. These metrics provide complementary insights into how different debiasing strategies, including re-ranking-based, causality-aware, and representation-driven methods, affect exposure patterns beyond pure relevance.

1) APLT

A higher *APLT* reflects the extent to which a debiasing method promotes long-tail items within the top- N recommendations, thereby encouraging exposure to less popular content. As illustrated in Figure 4, methods such as *MUL* and representation-oriented approaches including *DICE* and *CausE* achieve elevated *APLT* scores by aggressively prioritizing underrepresented items. While this strategy increases tail exposure, it may also introduce deviations from users’ historical popularity tendencies, a pattern consistent with their higher calibration errors observed in *Rmse-PC* and *MRMC*.

In contrast, recent optimization-driven frameworks such as *PDRO* and *PPAC* exhibit markedly low *APLT* values across backbone recommenders, indicating limited integration of tail items into the recommendation lists. This behavior suggests that distributional robustness or adaptive calibration objectives alone do not necessarily translate into effective long-tail promotion, and may instead preserve head-dominant recommendation structures.

Across all settings, *SUPER* achieves a balanced incorporation of tail items, maintaining competitive *APLT* without indiscriminately saturating recommendation lists with niche content. Rather than enforcing uniform corrective shifts, *SUPER* regulates tail exposure in accordance with each user’s popularity inclination, enabling meaningful discovery while preserving behavioral coherence.

This distinction becomes particularly important in scenarios characterized by heterogeneous user preferences, where overly aggressive tail promotion may undermine personalization, and insufficient tail exposure may limit recommendation

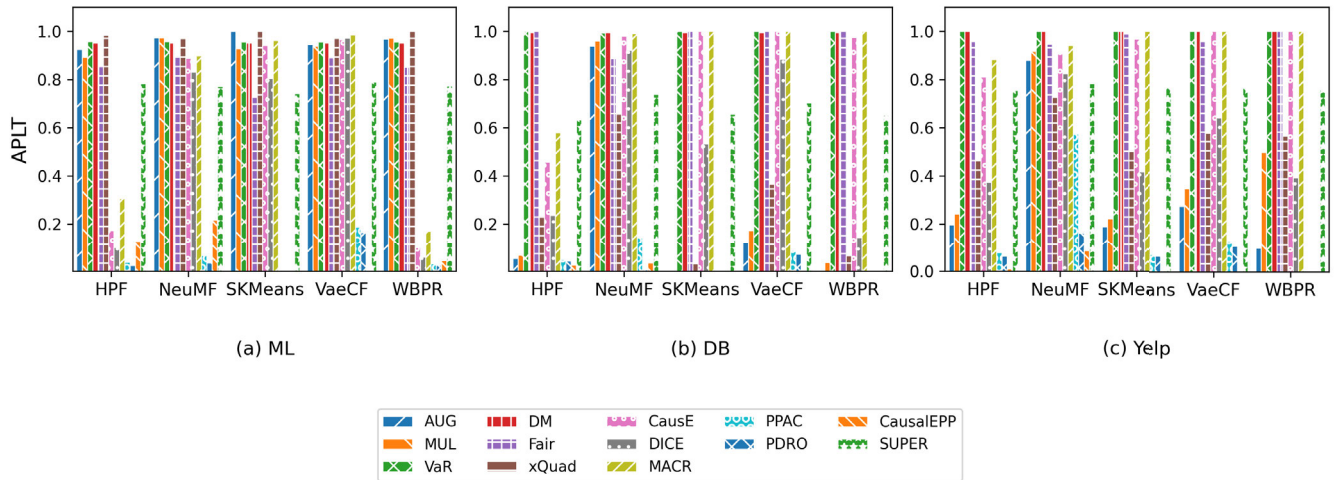


FIGURE 4. APLT results of the popularity-debiasing methods for the *HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR* algorithms on the (a) ML, (b) DB, and (c) Yelp datasets. Stars denote statistically significant improvements of *SUPER* over the second-best method and are displayed only when *SUPER* is the best-performing method; hence, no star means *SUPER* is not the best in that setting.

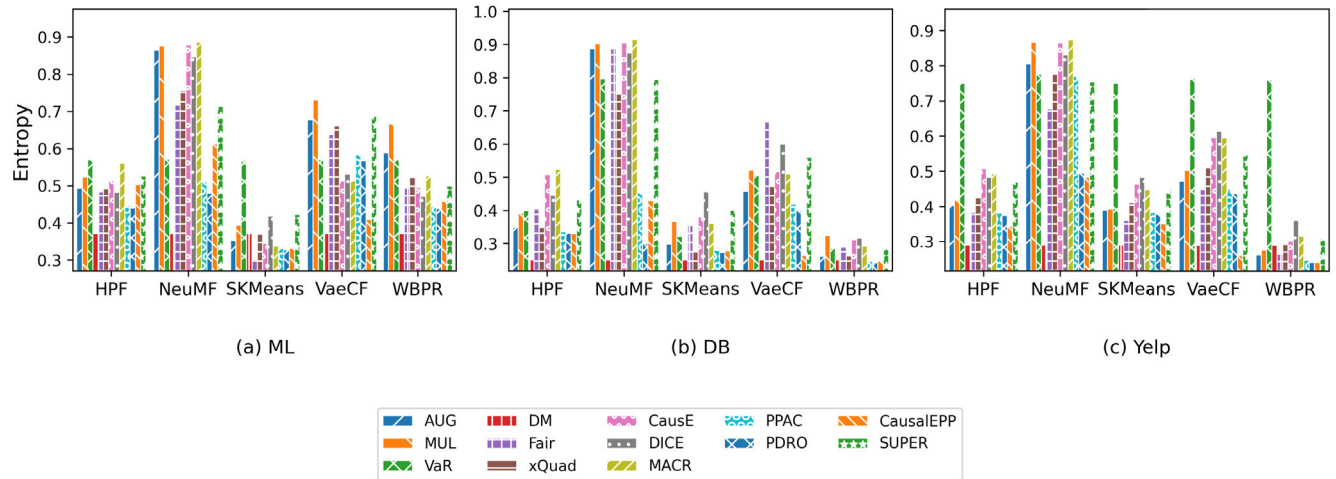


FIGURE 5. Entropy results of the popularity-debiasing methods for the *HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR* algorithms on the (a) ML, (b) DB, and (c) Yelp datasets. Stars denote statistically significant improvements of *SUPER* over the second-best method and are displayed only when *SUPER* is the best-performing method; hence, no star means *SUPER* is not the best in that setting.

novelty. By contrast, *SUPER* consistently navigates this trade-off, suggesting that user-aligned exposure control is critical for achieving sustainable long-tail integration.

Overall, these findings reinforce the central premise of this work: effective popularity debiasing is not solely a matter of maximizing tail exposure, but of calibrating that exposure in a manner consistent with user-specific consumption patterns.

2) ENTROPY

Entropy measures how evenly recommendations are distributed across the item catalog, reflecting overall diversity. As illustrated in Figure 5, approaches such as *xQuad*, *MUL*, and *MACR* frequently attain high entropy values by explicitly diversifying recommendation lists or enforcing balanced exposure across item groups. However, such gains are

often accompanied by weaker alignment with user-specific popularity tendencies, particularly when diversification is applied uniformly.

In addition, user-aware optimization and calibration frameworks such as *PDRO* and *PPAC*, as well as exposure-correction strategies like *CausalEPP*, tend to exhibit more moderate entropy behavior: they mitigate popularity effects through robustness, calibration, or exposure penalization objectives rather than aggressively maximizing catalog dispersion. As a result, their entropy gains are typically less pronounced than strongly diversification-driven re-ranking methods, but may be achieved with improved rank-level coherence.

By contrast, *SUPER* consistently attains competitive entropy across datasets without relying on rigid

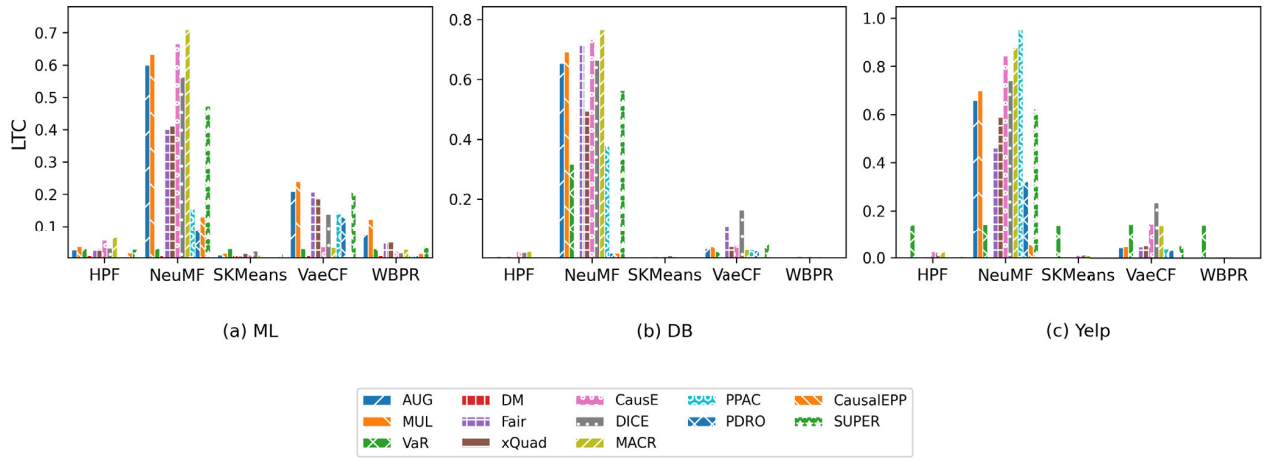


FIGURE 6. LTC results of the popularity-debiasing methods for the *HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR* algorithms on the (a) ML, (b) DB, and (c) Yelp datasets. Stars denote statistically significant improvements of *SUPER* over the second-best method and are displayed only when *SUPER* is the best-performing method; hence, no star means *SUPER* is not the best in that setting.

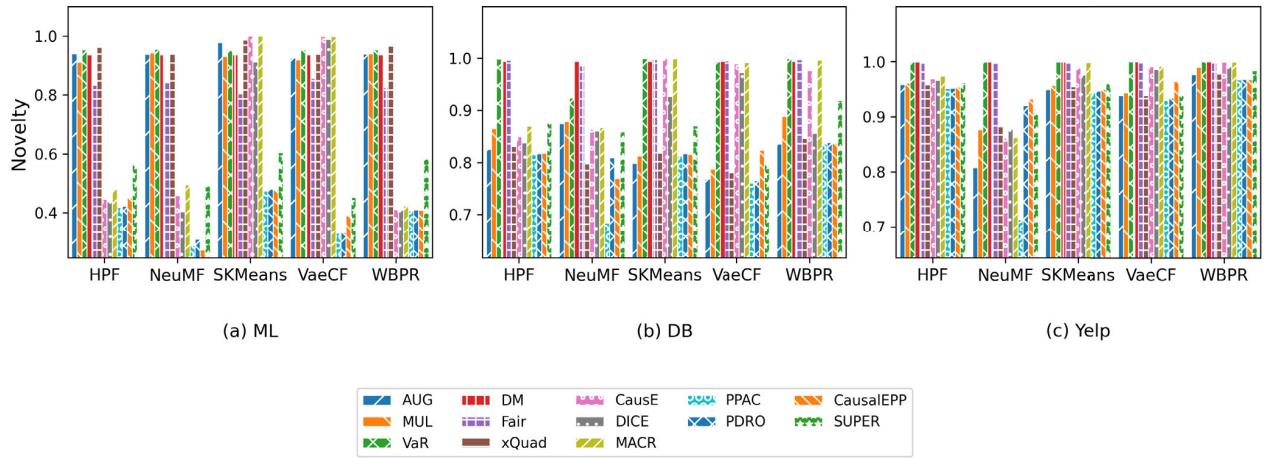


FIGURE 7. Novelty results of the popularity-debiasing methods for the *HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR* algorithms on the (a) ML, (b) DB, and (c) Yelp datasets. Stars denote statistically significant improvements of *SUPER* over the second-best method and are displayed only when *SUPER* is the best-performing method; hence, no star means *SUPER* is not the best in that setting.

diversification constraints. By dynamically balancing head and tail items according to each user's inferred inclination, *SUPER* preserves broad item exposure while maintaining strong popularity calibration, highlighting its ability to harmonize diversity and user-centric relevance.

3) LTC (LONG-TAIL COVERAGE)

While *APLT* captures the *per-user* fraction of tail items, *LTC* measures *system-level* coverage, i.e., how broadly tail items appear across all users' top-*N* lists. As shown in Figure 6, methods that aggressively promote tail content, such as *MUL* and *AUG*, as well as exposure-correction and representation/causality-oriented approaches like *CausE*, *DICE*, and *MACR*, tend to achieve higher *LTC* by spreading long-tail exposure across a larger portion of the catalog. A similar effect is observed for calibration-driven approaches such as *PPAC*, which can also yield strong tail coverage in some datasets by amplifying user-tail alignment.

However, maximizing *LTC* via broad tail promotion may come with the risk of overexposing niche items for popularity-oriented users, potentially weakening personalization and calibration. In contrast, *SUPER* achieves a consistently competitive (moderate-to-high) *LTC* across datasets by controlling tail exposure at the *user level* rather than enforcing a global objective. This user-centric exposure control preserves meaningful long-tail coverage without relying on blanket tail amplification. Methods such as *PDRO* and *CausalEPP*, which prioritize robustness or exposure-confounding correction, typically yield more conservative tail coverage, reflecting their different optimization focus.

4) NOVELTY

Novelty captures the extent to which a recommender introduces users to previously unseen or infrequently interacted items. As shown in Figure 7, methods that apply strong popularity suppression or aggressive long-tail promotion (e.g., *AUG*, *MUL*, and *DICE*) typically achieve the highest

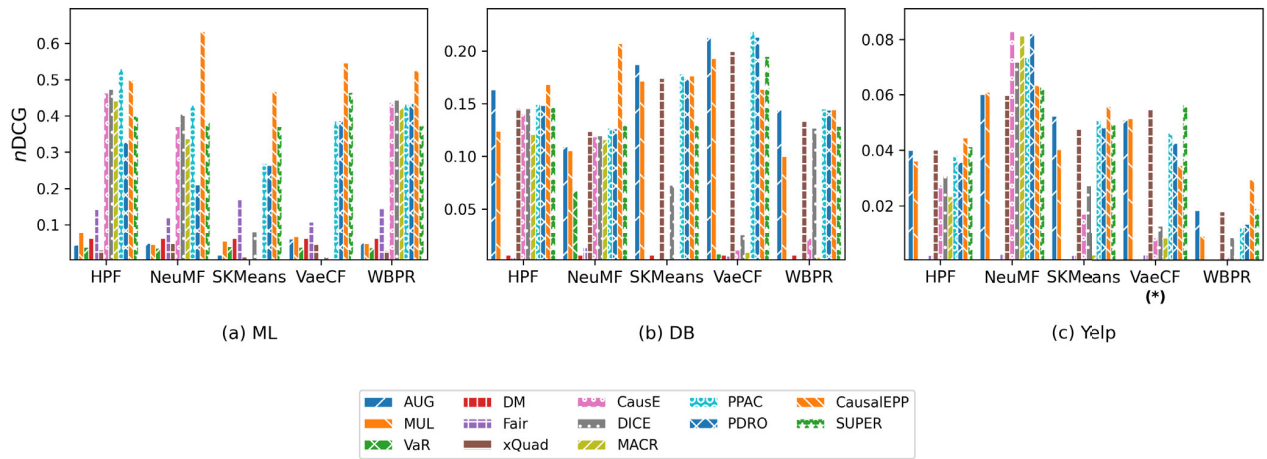


FIGURE 8. *nDCG* results of the popularity-debiasing methods for the *HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR* algorithms on the (a) *ML*, (b) *DB*, and (c) *Yelp* datasets. Stars denote paired Wilcoxon signed-rank significance across seeds for *SUPER* versus the second-best method, where (*), (**), and (***) indicate $p < 0.05$, $p < 0.01$, and $p < 0.001$. Stars are shown only when *SUPER* is the best-performing method; hence, no star means *SUPER* is not the best in that setting.

novelty, but this behavior can also coincide with weaker alignment to users' historical popularity tendencies (cf. *Rmse-PC* and *MRMC*).

Across datasets, *CausalEPP* is consistently competitive in novelty and can match or exceed the strongest baselines in several settings, reflecting its exposure-aware penalization of popularity-driven signals. *PDRO* and *PPAC* generally provide more moderate novelty gains, indicating a more conservative shift toward tail exploration.

Overall, *SUPER* yields consistently strong novelty without relying on indiscriminate exploration: by calibrating the head–tail composition at the user level and maintaining rank-level exposure coherence, it promotes meaningful discovery while preserving a stable and personalized recommendation structure.

C. RANKING ACCURACY: *nDCG*

Although the preceding analyses emphasize calibration and beyond-accuracy dimensions, ranking accuracy remains a fundamental determinant of recommendation quality. The *nDCG* (normalized Discounted Cumulative Gain) metric evaluates how effectively each method positions relevant items near the top of the ranked list. Figure 8 presents results for the *HPF*, *NeuMF*, *SKMeans*, *VaeCF*, and *WBPR* recommenders under a diverse set of popularity-debiasing strategies.

Several methods achieve strong ranking accuracy by prioritizing relevance preservation during debiasing. In particular, *CausalEPP* consistently delivers the highest *nDCG* scores, while calibration-oriented approaches such as *PPAC* also demonstrate competitive ranking performance. These results suggest that exposure-aware correction and personalized calibration can effectively mitigate popularity bias without severely disrupting relevance ordering.

Importantly, *SUPER* remains highly competitive in terms of *nDCG* across datasets and backbone recommenders,

despite simultaneously optimizing for calibration and beyond-accuracy objectives. Unlike methods that prioritize ranking accuracy through stronger exposure corrections, *SUPER* adopts a balanced strategy that regulates head–tail composition at the user level while preserving stable relevance signals.

This behavior highlights a key insight: maximizing ranking accuracy alone does not necessarily guarantee user-aligned recommendation quality. Methods that achieve the highest *nDCG* may still exhibit weaker calibration or less controlled exposure patterns (see Sections V-A and V-B). By contrast, *SUPER* demonstrates that strong ranking performance can coexist with improved popularity alignment and diversity.

Furthermore, the stability of *SUPER*'s *nDCG* across diverse base recommenders indicates that its benefits are not architecture-dependent. Even in challenging recommendation environments, the method maintains reliable relevance ordering while mitigating popularity bias. Collectively, these findings suggest that user-centric popularity calibration and accurate ranking are complementary rather than conflicting objectives.

These findings reinforce the central premise of this work: effective popularity debiasing should be evaluated not solely through ranking accuracy, but through its ability to balance relevance, calibration, and exposure.

D. HOLISTIC ANALYSIS: *GKPI*

Although individual metrics provide valuable insights into specific aspects of recommendation quality, interpreting overall system performance becomes challenging when these criteria yield conflicting signals. To address this issue, we employ a holistic evaluation measure, *GKPI*, which aggregates ranking accuracy and beyond-accuracy dimensions into a single composite score. Figure 9 presents

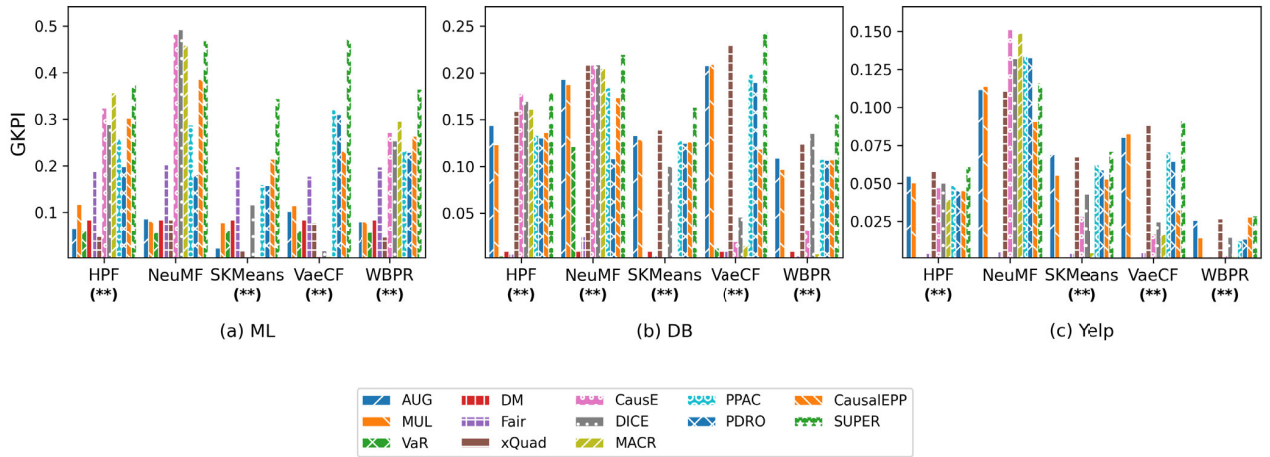


FIGURE 9. GKPI results of the popularity-debiasing methods for the HPF, NeuMF, SKMeans, VaeCF, and WBPR algorithms on the (a) ML, (b) DB, and (c) Yelp datasets. Stars denote paired Wilcoxon signed-rank significance across seeds for SUPER versus the second-best method, where (*), (**), and (***) indicate $p < 0.05$, $p < 0.01$, and $p < 0.001$. Stars are shown only when SUPER is the best-performing method; hence, no star means SUPER is not the best in that setting.

the resulting comparisons across all backbone recommenders and datasets.

From an integrated perspective, the compared methods exhibit distinct strengths across evaluation dimensions. Diversification-oriented strategies such as *MUL* and *xQuad* tend to improve catalog coverage and novelty-related indicators, whereas representation- and causality-driven approaches including *CausE*, *DICE*, and *MACR* emphasize exposure control and structural bias mitigation. User-aware optimization frameworks such as *PPAC* and robustness-oriented methods like *PDRO* further enhance calibration-oriented or stability-related metrics. These heterogeneous performance profiles highlight the inherently multi-objective nature of popularity debiasing.

In contrast, *SUPER* consistently achieves the highest GKPI across datasets and backbone recommenders, indicating a more balanced trade-off among competing recommendation objectives. Following our evaluation protocol, comparisons are reported against the second-best debiasing method (i.e., the strongest non-*SUPER* baseline under the seed-averaged mean for each setting). For example, on the *ML* dataset with the *VaeCF* recommender, *SUPER* attains a GKPI of 0.472, surpassing the runner-up method (*PPAC*, 0.321) by an absolute margin of 0.151.

These findings suggest that regulating head–tail exposure at the user level provides structural benefits that extend beyond isolated metric improvements. Rather than optimizing a single dimension at the expense of others, *SUPER* maintains stable performance across accuracy, calibration, and beyond-accuracy criteria, which is particularly desirable in real-world recommendation environments where multiple quality dimensions must be satisfied simultaneously. This behavior supports the interpretation of GKPI as a practical rubric for identifying methods that achieve coherent trade-offs rather than extreme optimization of isolated metrics.

E. RUNTIME ANALYSIS: TOP-N RECOMMENDATION GENERATION COST

Beyond recommendation quality and calibration, computational efficiency is a critical factor for the practical deployment of popularity-debiasing methods. We therefore analyze the average wall-clock time required to generate top-10 recommendation lists under different debiasing strategies. The reported values correspond to the total post-processing cost, including candidate selection and list construction, averaged across all backbone recommenders and random seeds. For each dataset and method, we report the mean wall-clock time aggregated over all backbone recommenders and seeds (i.e., averaging the per-(recommender, seed) runtimes).

All experiments were conducted on a single machine equipped with an Apple M2 Max processor (12-core CPU, 38-core GPU), 32 GB RAM, and 1 TB SSD storage. No additional model retraining was performed during top- N generation; all methods operate purely at the inference or post-processing stage. The reported times measure only inference-time ranking and post-processing during top- N generation. They *exclude* all training costs; for *SUPER*, the two backbone models (head-only and tail-only) are trained offline with the same architecture and hyperparameters, and runtime reflects only the final merging/calibration step.

Table 3 highlights substantial differences in computational cost among debiasing strategies. Network-flow-based methods such as Discrepancy Minimization (DM) incur the highest runtime overhead, reflecting their inherent optimization complexity. Similarly, diversification-oriented approaches such as VarERPs (which subsume AUG, MUL, and VaR) and *xQuad* require repeated score recomputation and iterative re-ranking, leading to significantly higher execution times.

In contrast, *SUPER* consistently exhibits runtime performance comparable to lightweight post-processing methods such as *CausE*, *DICE*, and *MACR*, and remains close to the BASE recommendation cost. Across all datasets, the

TABLE 3. Average wall-clock time (in seconds) to generate Top-10 recommendation lists for *all* users in each dataset, including ranking and post-processing, averaged over backbone recommenders and random seeds.

Popularity-Debiasing Benchmarks	ML	DoubanBooks	Yelp
Backbone (No Debiasing)	10.17	132.56	99.42
VarERPs (AUG/MUL/VaR)	406.61	5101.83	4065.30
DM	5013.48	10430.10	10866.38
FA*IR	49.46	644.72	514.59
xQuad	406.37	5088.11	4083.80
CausE	13.04	172.28	137.32
DICE	13.43	176.03	141.42
MACR	12.98	170.46	137.03
PDRO	14.97	178.69	107.81
PPAC	1462.82	8566.27	1680.84
CausalEPP	283.21	2783.36	2007.03
SUPER	11.11	146.34	103.09

additional overhead introduced by *SUPER* remains marginal, despite its user-specific popularity calibration and iterative blueprint-based merging mechanism. This efficiency stems from the fact that *SUPER* avoids costly global optimization procedures and does not require additional model training during inference.

Overall, these results demonstrate that *SUPER* achieves a favorable balance between calibration quality, beyond-accuracy improvements, and computational efficiency, making it well suited for large-scale recommendation pipelines where both effectiveness and runtime constraints are critical.

While the runtime analysis confirms that *SUPER* remains computationally lightweight under offline evaluation settings, the following section discusses remaining limitations related to dynamic popularity shifts, cold-start conditions, and large-scale deployment considerations.

F. POPULARITY-THRESHOLD SENSITIVITY

We analyzed the sensitivity of *SUPER* to the head–tail popularity threshold by evaluating multiple head thresholds (10%, 20%, 30%, 40%, and 50%) across all datasets and backbone recommenders. The results reveal clear trends across metrics. Lower thresholds may slightly improve calibration (RMSE-PC and MRMC) by adopting a stricter notion of popularity, whereas larger thresholds generally weaken calibration and reduce the overall balance among objectives. When jointly considering accuracy, calibration, and beyond-accuracy metrics through GKPI, the 20% threshold consistently provides the most robust trade-off, achieving the highest average GKPI across datasets and serving as the best-performing choice in the majority of configurations. While a small number of dataset–backbone combinations favor alternative thresholds, these deviations are limited and do not undermine the overall stability of the 20% setting. Accordingly, we adopt 20% as a stable default for comparability, while noting that the threshold can be selected in a fully data-driven manner by maximizing GKPI on a validation split.

G. COMPONENT-WISE ABLATION OF SUPER

To better understand the contribution of each component in *SUPER*, we conducted an ablation study with two

variants. The full *SUPER* model combines Pareto-based head–tail partitioning, a user-specific popularity-inclination ratio Pop_u , and the blueprint-based merge. In *Variant A*, we removed personalization by replacing Pop_u with a fixed 50/50 head–tail split for all users, while keeping both the Pareto partition and the blueprint unchanged. In *Variant B*, we retained the Pareto partition and the user-specific Pop_u , but replaced the blueprint with a random merge that only preserves the overall head–tail ratio in expectation.

Across all datasets and baseline recommenders, *Variant A* consistently exhibits higher $Rmse-PC$ than *SUPER*, with the average calibration error increasing from approximately 0.035 to 0.065, and a slight decrease in the average GKPI, indicating that the user-specific Pop_u is the main driver of *SUPER*'s calibration gains. In contrast, *Variant B* yields almost identical $Rmse-PC$ to *SUPER* but slightly lower GKPI on average, suggesting that the blueprint-based merge mainly contributes to improving the holistic trade-off captured by GKPI rather than to calibration alone. Since all three configurations share the same Pareto-based partitioning, its effect is primarily reflected in the improvements of *SUPER* (and its variants) over the non-debiased baselines reported in the main experiments. These trends were consistent across all datasets and backbone recommenders; for brevity, we do not report the full ablation table.

Overall, these ablation results empirically validate the design of *SUPER* by showing that both user-specific popularity inclination and blueprint-based merging are necessary to achieve strong calibration and a balanced holistic performance.

H. USER SEGMENTS AND DATASET CHARACTERISTICS

To further understand the behavior of *SUPER* under different conditions, we analyze its performance across user segments and dataset characteristics, including popularity-oriented versus tail-oriented users, sparse versus dense datasets, and cold-start regimes.

1) POPULARITY-ORIENTED VERSUS TAIL-ORIENTED USERS

Regarding user popularity orientation, *SUPER* exhibits adaptive behavior across the preference spectrum. Users with a strong inclination toward popular (head) items benefit from

SUPER's calibrated exposure mechanism, which prevents excessive injection of tail content that could undermine relevance. Conversely, users who historically favor tail items receive a higher proportion of niche recommendations without completely excluding popular items, ensuring that exploration is encouraged while maintaining recommendation coherence. This observation confirms that *SUPER* does not enforce a uniform head–tail balance, but instead dynamically aligns recommendation exposure with individual user tendencies.

2) DATASET SPARSITY AND DOMAIN CHARACTERISTICS

Dataset characteristics further influence this behavior. In denser datasets such as MovieLens-1M, user popularity inclination can be estimated more reliably, resulting in particularly strong calibration and stable ranking behavior. In contrast, highly sparse datasets such as Yelp pose greater challenges due to limited interaction histories. Despite this, *SUPER* remains robust and competitive, indicating that it does not rely on overly precise popularity estimates and can function effectively even under severe sparsity.

3) USER ACTIVITY LEVELS AND COLD-START ROBUSTNESS

To further examine robustness with respect to user activity, we stratified users into three groups according to their total number of interactions: low-activity (<20), medium-activity (20–100), and high-activity (>100). For each group, we re-evaluated all debiasing methods across all datasets and baseline recommenders.

The results for low-activity (cold-start) users indicate that Yelp is largely dominated by sparse profiles (13,685 out of 16,239 users; 84.27% have < 20 interactions), where calibration is inherently challenging due to limited behavioral evidence. In this regime, RMSE-PC values remain small in absolute magnitude, yet clear relative differences still emerge across debiasing strategies. Notably, *SUPER* achieves the lowest average RMSE-PC of 0.0096 across backbone recommenders, outperforming strong baselines such as PDRO (0.0234), PPAC (0.0294), DICE (0.0435), and CausalePP (0.0254). Even compared to competitive calibration-oriented methods like Fair (0.0249) and CausE (0.0377), *SUPER* maintains the best calibration, providing quantitative evidence of robustness under sparse interaction histories. For medium- and high-activity users, where the user-specific popularity inclination Pop_u can be estimated more reliably, *SUPER* remains among the top-performing methods in calibration while preserving competitive accuracy and beyond-accuracy performance.

α : USER-CENTERED FAIRNESS VS. SYSTEM-LEVEL EXPOSURE

While metrics such as Entropy, LTC, and Gini-type measures capture how evenly recommendation exposure is distributed across items at the system level, they do not directly reflect fairness at the individual user level. In this work, user-centered fairness is primarily evaluated through calibration-oriented metrics (Rmse-PC and MRMC), which

quantify how well recommendation exposure aligns with each user's historical popularity inclination.

Taken together with the above segment analyses, the results suggest that *SUPER* adapts smoothly across diverse user profiles without introducing systematic disadvantages to any particular group. Rather than enforcing uniform exposure across users, *SUPER* prioritizes aligning recommendation exposure with individual user tendencies, thereby providing a user-centric notion of fairness that remains robust under heterogeneity in both user activity and dataset sparsity.

I. DISCUSSION OF OVERALL PERFORMANCE

Taken collectively, the experimental results across calibration, beyond-accuracy, and ranking accuracy dimensions demonstrate that *SUPER* not only aligns recommended content with users' popularity preferences, but also consistently enhances diversity, long-tail exposure, and novel item discovery. While several debiasing techniques outperform *SUPER* in isolated metrics, such as *MUL* in terms of *APLT*, *xQuad* with respect to *Entropy*, or causality- and representation-driven methods like *CausE* and *DICE* in catalog-level exposure, these gains are frequently accompanied by increased calibration error or instability across ranked positions.

Methods that rely on strong global constraints or uniform corrective mechanisms often overshoot the intended proportion of tail items, leading to misalignment with individual users' historical popularity inclinations. In particular, approaches such as *CausE*, *DICE*, and *MACR* may effectively mitigate exposure bias at an aggregate level, yet struggle to preserve rank-level coherence or user-specific balance, especially in sparse or heterogeneous datasets. Conversely, purely accuracy-driven baselines tend to maintain relevance but offer limited diversity, novelty, and long-tail coverage.

By coupling user-specific popularity inclination with a dynamically balanced item selection strategy, *SUPER* adapts robustly to both moderate and sparse data conditions. This adaptability benefits not only calibration-oriented measures (*Rmse-PC*, *MRMC*), but also promotes equitable item exposure and meaningful exploration without resorting to indiscriminate tail promotion. As a result, *SUPER* achieves a more favorable overall trade-off across accuracy, fairness, and diversity objectives. More broadly, these findings suggest that popularity debiasing should not be framed solely as a diversity-enhancement or exposure-correction task, but rather as a structured calibration problem that requires balancing relevance, exposure, and user-specific consumption patterns. The consistent performance of *SUPER* indicates that regulating the positional structure of exposure, rather than relying exclusively on global optimization objectives, can produce systemic improvements across multiple recommendation dimensions. This perspective highlights exposure structure as a controllable design variable in recommender systems, rather than a byproduct of ranking optimization, and underscores the importance of rank-aware calibration for achieving user-aligned recommendation quality.

In practical terms, these findings position *SUPER* as a robust and versatile debiasing framework for real-world recommendation systems. Rather than optimizing a single dimension in isolation, *SUPER* delivers stable and well-balanced performance across multiple evaluation criteria, ensuring that personalization, diversity, and alignment with users' observed popularity tendencies are jointly preserved.

Finally, all evaluations are conducted under a static offline setting using fixed historical interaction data, which is the standard protocol adopted in popularity-bias and calibration research; consequently, dynamic popularity adaptation and real-time preference drift are intentionally outside the scope of this study. Future work may therefore explore adaptive exposure regulation under dynamic interaction environments.

Additional Remarks:

The popularity inclination proxy (Pop_u) in *SUPER* is computed from the user's full interaction history to capture overall exposure tendencies toward popular versus tail items. Evaluation metrics such as RMSE-PC, on the other hand, focus on positively perceived items. This distinction allows the framework to model exposure calibration while preserving preference-oriented evaluation.

The blueprint used in the merging stage is derived from preference-ranked interactions rather than chronological order. This design aims to provide a stable representation of user preference structure, especially in public datasets where timestamp information may be incomplete or inconsistent.

Finally, the GKPI indicator is intended as a compact comparative summary of multiple evaluation dimensions within the same experimental setting. Individual metrics remain the primary basis for interpretation, while GKPI highlights the overall balance between accuracy and long-tail exposure.

J. KEY OBSERVATIONS AND IMPLICATIONS

The experimental findings collectively emphasize that popularity debiasing is most effective when treated as a user-centric calibration problem rather than a purely system-level exposure optimization task. Instead of enforcing uniform head–tail ratios or aggressive long-tail promotion across all users, *SUPER* demonstrates that aligning recommendation exposure with individual popularity inclinations leads to a more stable and interpretable trade-off among accuracy, diversity, and fairness objectives.

A key implication of these results is that strong calibration does not necessarily conflict with beyond-accuracy goals. By dynamically adjusting the head–tail composition during list construction, *SUPER* achieves meaningful improvements in long-tail exposure, diversity, and novelty while preserving ranking quality and avoiding overcorrection. This behavior contrasts with several competing debiasing strategies that optimize a single objective at the expense of user-level coherence.

From a practical perspective, these observations suggest that future popularity-debiasing approaches should

incorporate personalization as a first-class component rather than relying on global constraints or static re-ranking rules. *SUPER* provides a scalable example of how user-specific preference modeling can be integrated into post-processing pipelines without incurring substantial computational overhead.

Overall, the results indicate that user-aligned popularity calibration offers a principled and robust pathway toward fairer and more diverse recommender systems, particularly in real-world settings characterized by heterogeneous user preferences and data sparsity.

VI. LIMITATIONS

Although *SUPER* demonstrates promising effectiveness in mitigating popularity bias and maintaining robust calibration under controlled experimental settings, several limitations emerge when one considers its applicability in more complex, real-world scenarios.

A. LIMITATIONS RELATED TO POPULARITY MODELING

1) POPULARITY INCLINATION AS A SINGLE PREFERENCE DIMENSION

We acknowledge that the popularity inclination ratio (Pop_u) captures only one dimension of user preference and does not aim to represent the full spectrum of individual tastes. In *SUPER*, popularity inclination is intentionally modeled as an orthogonal control signal that regulates exposure to head versus tail items, while all other preference dimensions (e.g., genre affinity, topical relevance, or rating intensity) remain fully governed by the underlying backbone recommender. In this sense, *SUPER* does not replace or override standard preference modeling, but complements it by calibrating popularity exposure at the user level.

Furthermore, since the public benchmark datasets used in this study do not provide impression or exposure logs, it is not possible to fully disentangle preference from exposure in a causal sense. *SUPER* therefore treats popularity inclination as a robust behavioral estimate rather than a true latent preference. We further emphasize that popularity debiasing inherently involves normative design choices regarding how exposure should be regulated. *SUPER* makes these assumptions explicit rather than implicit by framing popularity inclination as an operational calibration signal rather than a definitive representation of true user preference.

2) LIMITS OF CAUSAL PREFERENCE IDENTIFICATION

We emphasize that *SUPER* is not designed as a causal inference framework and does not attempt to recover users' latent "true preferences" from observational interaction data. Disentangling preference from exposure typically requires impression logs, randomized interventions, or strong identification assumptions that are not available in standard offline recommendation benchmarks. Accordingly, *SUPER* adopts a pragmatic perspective by treating observed interactions as behavioral signals subject to exposure effects and

focuses on regulating how these signals influence ranking outcomes. The proposed framework therefore operates as an exposure-calibrated ranking mechanism rather than a deconfounding model. Investigating causal extensions that integrate propensity estimation or counterfactual learning represents an important direction for future work.

Similarly, *SUPER* is not designed as a time-aware recommender; instead, it functions as a model-agnostic exposure-calibration layer that can operate on top of backbone algorithms with or without temporal dynamics.

3) DYNAMICS OF POPULARITY AND USER PREFERENCE DRIFT

The current formulation of *SUPER* estimates item popularity and user inclination under a static offline setting. However, the framework naturally supports dynamic updates with minimal modification. Item popularity statistics can be recomputed periodically or within sliding temporal windows to reflect evolving consumption trends. Similarly, the user-specific popularity inclination Pop_u can be updated incrementally using recent interactions, enabling the system to adapt when a user's preference shifts toward more popular or more niche content over time. Importantly, such updates do not require retraining the underlying recommendation models; only the lightweight popularity statistics and post-processing ratios need to be refreshed. This design makes *SUPER* compatible with drift-aware and real-time recommendation pipelines, although a full temporal evaluation is left for future work.

B. DATA, SCALABILITY, AND DEPLOYMENT CONSTRAINTS

Another limitation arises from the system's data requirements. By design, *SUPER* depends on meaningful historical user interactions to gauge how strongly individuals lean toward popular or less-known items. This requirement poses challenges in cold-start conditions where new users or newly introduced products lack sufficient interaction logs. Under these constraints, calibrating head–tail preferences becomes less reliable, and the model may offer suboptimal suggestions until enough data accumulates.

Further considerations pertain to the computational overhead associated with integrating user-centric popularity estimation into each recommendation. While the hybrid pre- and post-processing approach in *SUPER* is generally feasible for moderate data scales, very large systems with millions of active users and items might demand further optimization. Techniques such as incremental updates to the popularity classification or parallelized operations could mitigate performance bottlenecks, but the extent of these modifications remains to be assessed.

Additionally, privacy-oriented constraints can limit the availability of detailed user interaction data. As *SUPER* relies on clear signals of user preferences and item popularity, approaches that blur or anonymize these inputs may reduce its precision in distinguishing head and tail items or in estimating

a user's observed inclination for mainstream versus niche content.

C. ASSUMPTION SENSITIVITY AND POTENTIAL NEGATIVE EFFECTS

1) SENSITIVITY TO THE PARETO THRESHOLD

Although the 20% head threshold provides stable performance across datasets, the definition of popularity directly influences the estimation of user inclination and the resulting exposure calibration. Lower thresholds impose a stricter notion of popularity and may improve calibration, whereas higher thresholds can blur the distinction between head and tail items, potentially weakening exposure control. This sensitivity highlights that popularity partitioning is not purely structural but also assumption-dependent.

2) FAILURE MODES OF THE BLUEPRINT STRATEGY

The blueprint-based merge uses a user's historical head–tail pattern as a soft positional prior for interleaving popular and tail candidates. For users with sparse, short, or noisy histories, this pattern may be unreliable and can inject incidental exposure noise into rank-level ordering. Notably, *SUPER*'s main calibration behavior is governed by the hard head–tail quota derived from Pop_u , while the blueprint only affects within-list placement as long as quota remains. Therefore, in cold-start or low-activity regimes, a robust variant can downweight or disable the blueprint and rely on proportional (quota-preserving) interleaving, optionally with simple smoothing based on history length.

3) REPRESENTATION FRAGMENTATION FROM INDEPENDENT TRAINING

Training separate models on head and tail subsets improves exposure controllability but may reduce cross-subset knowledge transfer. Consequently, latent signals learned in the popular domain may not fully inform tail-item representations, potentially limiting recommendation quality for niche items. Joint or partially shared representations could mitigate this effect but may simultaneously reintroduce popularity bias.

4) PERSONALIZATION-DIVERSITY TRADE-OFF

While *SUPER* aims to calibrate exposure rather than maximize diversity, increasing tail representation may not benefit all users equally. Popularity-oriented users may experience slight relevance degradation when additional tail items are introduced, whereas tail-oriented users typically benefit from improved discovery. This reflects a fundamental trade-off in popularity debiasing between exposure fairness and strict personalization.

VII. CONCLUSION AND FUTURE DIRECTIONS

This study introduced *SUPER*, a hybrid user-centric framework designed to mitigate popularity bias by calibrating recommendation exposure according to each user's observed inclination toward popular versus tail content. By combining Pareto-based popularity profiling, specialized

dual modeling, and a blueprint-guided calibration strategy, the proposed approach integrates head and tail items while preserving personalization. Extensive experiments across multiple datasets and backbone recommenders demonstrate that *SUPER* maintains strong ranking performance, achieves robust popularity alignment, and supports beyond-accuracy objectives without relying on strong causal assumptions. These findings highlight the feasibility of balancing exposure and relevance through lightweight calibration under standard offline evaluation settings.

Several directions remain for future research. Incorporating temporal dynamics and drift-aware updates may further improve adaptability in rapidly evolving platforms, while enhanced initialization strategies could strengthen performance under sparse interaction regimes. Additionally, extending the framework to privacy-preserving or federated environments represents an important step toward real-world deployment.

Overall, *SUPER* provides a scalable and model-agnostic pathway toward more balanced and user-aligned recommender systems, supporting the broader objective of delivering fair, diverse, and behaviorally coherent recommendations.

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DATA AVAILABILITY STATEMENT

The data will be made available upon reasonable request.

DECLARATION OF GENERATIVE AI IN SCIENTIFIC WRITING

During the preparation of this work the authors used ChatGPT in order to refine written text for grammar checking, readability, and fluency. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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